

```

# Импорт библиотек
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

# Отключаем предупреждения для чистоты вывода
warnings.filterwarnings('ignore')

# Настройка стиля графиков seaborn
sns.set(style="whitegrid")

# Загрузка данных
data = pd.read_csv('V8_classification_lr3.csv')

# Выводим первые 5 строк для первичного ознакомления с данными
display(data.head())

```

	0	1	2	3	4	5	6	7	8	9	...
48 \											
0	0.00	0.64	0.64	0.0	0.32	0.00	0.00	0.00	0.00	0.00	...
0.00											
1	0.21	0.28	0.50	0.0	0.14	0.28	0.21	0.07	0.00	0.94	...
0.00											
2	0.06	0.00	0.71	0.0	1.23	0.19	0.19	0.12	0.64	0.25	...
0.01											
3	0.00	0.00	0.00	0.0	0.63	0.00	0.31	0.63	0.31	0.63	...
0.00											
4	0.00	0.00	0.00	0.0	0.63	0.00	0.31	0.63	0.31	0.63	...
0.00											

	49	50	51	52	53	54	55	56	Class
0	0.000	0.0	0.778	0.000	0.000	3.756	61	278	1
1	0.132	0.0	0.372	0.180	0.048	5.114	101	1028	1
2	0.143	0.0	0.276	0.184	0.010	9.821	485	2259	1
3	0.137	0.0	0.137	0.000	0.000	3.537	40	191	1
4	0.135	0.0	0.135	0.000	0.000	3.537	40	191	1

[5 rows x 58 columns]

```

# а. Ознакомление с данными с помощью описательной статистики
print("Информация о наборе данных:")
display(data.info())

```

```

Информация о наборе данных:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4601 entries, 0 to 4600
Data columns (total 58 columns):
#   Column  Non-Null Count  Dtype

```

---	-----	-----	-----
0	0	4601 non-null	float64
1	1	4601 non-null	float64
2	2	4601 non-null	float64
3	3	4601 non-null	float64
4	4	4601 non-null	float64
5	5	4601 non-null	float64
6	6	4601 non-null	float64
7	7	4601 non-null	float64
8	8	4601 non-null	float64
9	9	4601 non-null	float64
10	10	4601 non-null	float64
11	11	4601 non-null	float64
12	12	4601 non-null	float64
13	13	4601 non-null	float64
14	14	4601 non-null	float64
15	15	4601 non-null	float64
16	16	4601 non-null	float64
17	17	4601 non-null	float64
18	18	4601 non-null	float64
19	19	4601 non-null	float64
20	20	4601 non-null	float64
21	21	4601 non-null	float64
22	22	4601 non-null	float64
23	23	4601 non-null	float64
24	24	4601 non-null	float64
25	25	4601 non-null	float64
26	26	4601 non-null	float64
27	27	4601 non-null	float64
28	28	4601 non-null	float64
29	29	4601 non-null	float64
30	30	4601 non-null	float64
31	31	4601 non-null	float64
32	32	4601 non-null	float64
33	33	4601 non-null	float64
34	34	4601 non-null	float64
35	35	4601 non-null	float64
36	36	4601 non-null	float64
37	37	4601 non-null	float64
38	38	4601 non-null	float64
39	39	4601 non-null	float64
40	40	4601 non-null	float64
41	41	4601 non-null	float64
42	42	4601 non-null	float64
43	43	4601 non-null	float64
44	44	4601 non-null	float64
45	45	4601 non-null	float64
46	46	4601 non-null	float64
47	47	4601 non-null	float64

```

48 48      4601 non-null float64
49 49      4601 non-null float64
50 50      4601 non-null float64
51 51      4601 non-null float64
52 52      4601 non-null float64
53 53      4601 non-null float64
54 54      4601 non-null float64
55 55      4601 non-null int64
56 56      4601 non-null int64
57 Class   4601 non-null int64

```

```
dtypes: float64(55), int64(3)
```

```
memory usage: 2.0 MB
```

```
None
```

```

print("\nОписательная статистика числовых признаков:")
display(data.describe())

```

```
Описательная статистика числовых признаков:
```

	0	1	2	3	4
\					
count	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000
mean	0.104553	0.213015	0.280656	0.065425	0.312223
std	0.305358	1.290575	0.504143	1.395151	0.672513
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.420000	0.000000	0.380000
max	4.540000	14.280000	5.100000	42.810000	10.000000

	5	6	7	8	9
...					
\					
count	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000
...					
mean	0.095901	0.114208	0.105295	0.090067	0.239413
...					
std	0.273824	0.391441	0.401071	0.278616	0.644755
...					
min	0.000000	0.000000	0.000000	0.000000	0.000000
...					
25%	0.000000	0.000000	0.000000	0.000000	0.000000

```

...
50%      0.000000      0.000000      0.000000      0.000000      0.000000
...
75%      0.000000      0.000000      0.000000      0.000000      0.160000
...
max      5.880000      7.270000      11.110000      5.260000      18.180000
...

          48          49          50          51          52
\
count  4601.000000  4601.000000  4601.000000  4601.000000  4601.000000
mean    0.038575    0.139030    0.016976    0.269071    0.075811
std     0.243471    0.270355    0.109394    0.815672    0.245882
min     0.000000    0.000000    0.000000    0.000000    0.000000
25%     0.000000    0.000000    0.000000    0.000000    0.000000
50%     0.000000    0.065000    0.000000    0.000000    0.000000
75%     0.000000    0.188000    0.000000    0.315000    0.052000
max     4.385000    9.752000    4.081000    32.478000    6.003000

          53          54          55          56
Class
count  4601.000000  4601.000000  4601.000000  4601.000000
4601.000000
mean    0.044238    5.191515    52.172789    283.289285
0.394045
std     0.429342    31.729449    194.891310    606.347851
0.488698
min     0.000000    1.000000    1.000000    1.000000
0.000000
25%     0.000000    1.588000    6.000000    35.000000
0.000000
50%     0.000000    2.276000    15.000000    95.000000
0.000000
75%     0.000000    3.706000    43.000000    266.000000
1.000000
max     19.829000   1102.500000   9989.000000   15841.000000
1.000000

```

```
[8 rows x 58 columns]
```

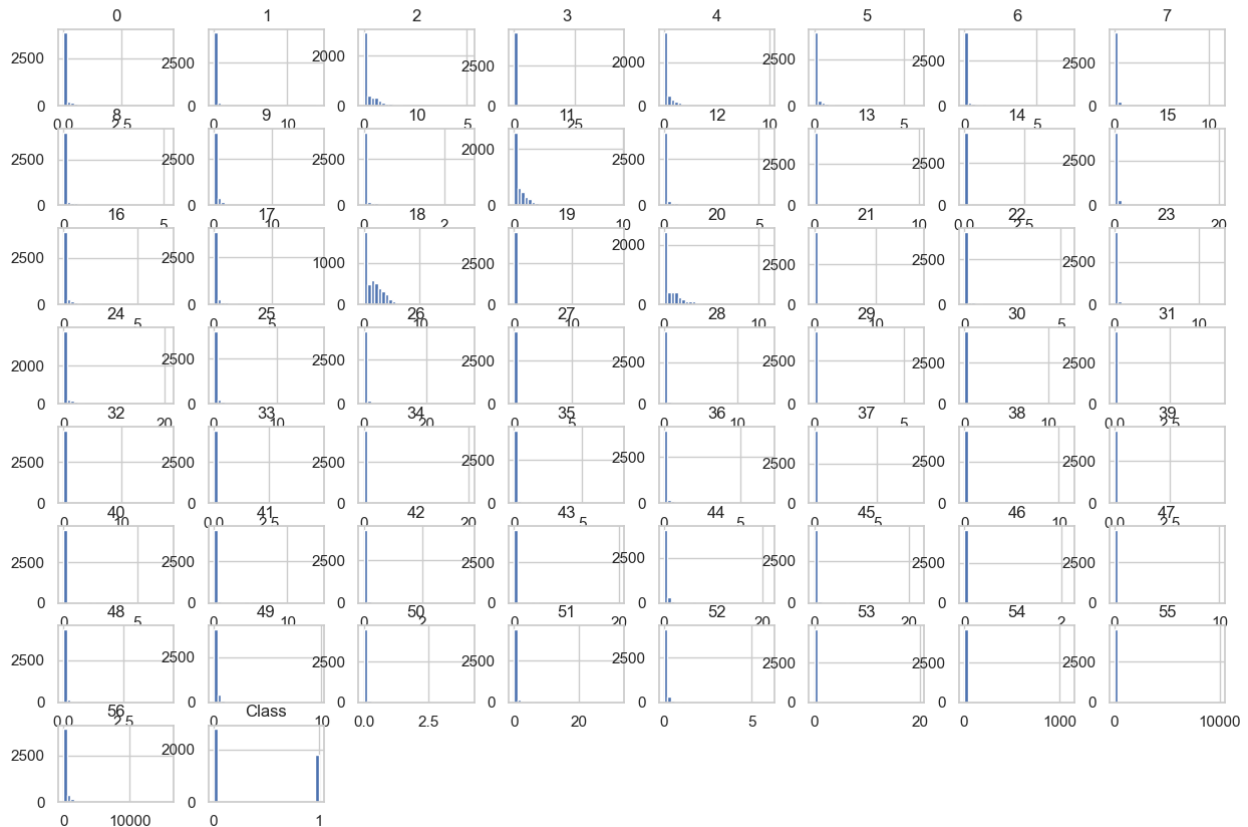
```
# b. Визуализация данных
```

```
# 1. Одномерная визуализация: построим гистограммы для каждого
```

числового признака

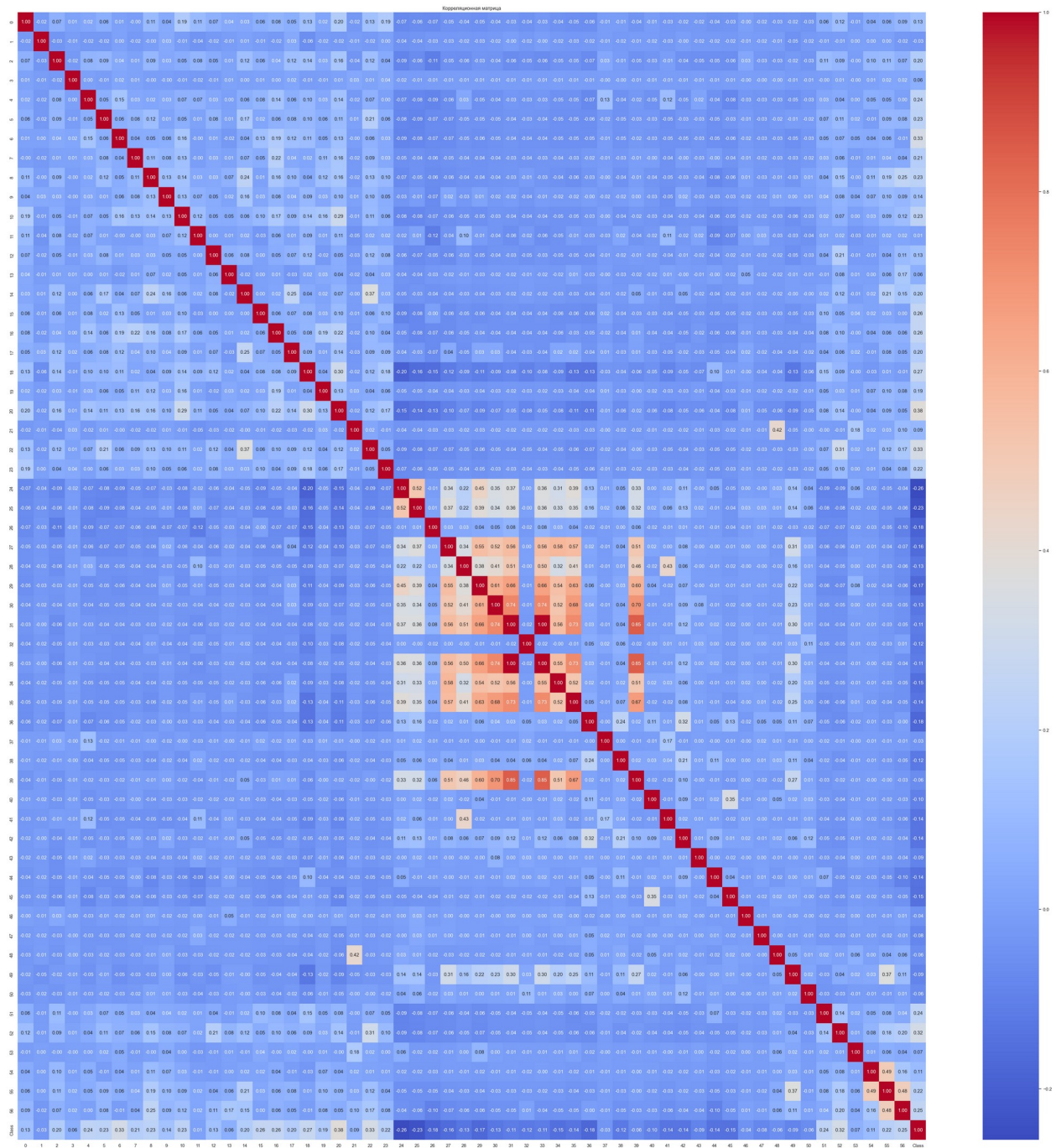
```
data.hist(bins=30, figsize=(15, 10))  
plt.suptitle("Гистограммы распределения признаков", fontsize=16)  
plt.show()
```

Гистограммы распределения признаков



2. Многомерная визуализация: тепловая карта корреляций между признаками

```
plt.figure(figsize=(50, 50))  
corr_matrix = data.corr()  
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm")  
plt.title("Корреляционная матрица")  
plt.show()
```



```
corr_matrix["Class"].sort_values(ascending=False)
```

```
Class    1.000000
20       0.383234
22       0.334787
6         0.332117
52        0.323629
18        0.273651
15        0.263215
```

16	0.263204
56	0.249164
4	0.241920
51	0.241888
10	0.234529
5	0.232604
8	0.231551
23	0.216111
55	0.216097
7	0.206808
17	0.204208
2	0.196988
14	0.195902
19	0.189761
9	0.138962
12	0.132927
0	0.126208
54	0.109999
21	0.091860
53	0.065067
13	0.060027
3	0.057371
11	0.007741
1	-0.030224
37	-0.031035
46	-0.044679
48	-0.059630
50	-0.064709
39	-0.064801
47	-0.084020
49	-0.089672
43	-0.094594
40	-0.097375
33	-0.112754
31	-0.114214
32	-0.119931
38	-0.122831
30	-0.126912
28	-0.133523
42	-0.135664
35	-0.136134
41	-0.136615
44	-0.140408
45	-0.146138
34	-0.149225
27	-0.158800
29	-0.171095
36	-0.178045
26	-0.183404

```

25      -0.232968
24      -0.256723
Name: Class, dtype: float64

data_changed = data.drop(["33", "21", "53", "13", "3", "11", "1",
"37", "46", "48", "50", "39", "47", "49", "43", "40"], axis=1,
inplace=False)

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

# Стандартизация исходного набора данных (data)
# Исключаем целевой признак 'Class'
features = data.drop("Class", axis=1).columns
data_scaled_features = scaler.fit_transform(data[features])
data_scaled = pd.DataFrame(data_scaled_features, columns=features)
data_scaled["Class"] = data["Class"].values

# Стандартизация изменённого набора данных (data_changed)
# Исключаем целевой признак 'Class'
features_changed = data_changed.drop("Class", axis=1).columns
data_changed_scaled_features =
scaler.fit_transform(data_changed[features_changed])
data_changed_scaled = pd.DataFrame(data_changed_scaled_features,
columns=features_changed)
data_changed_scaled["Class"] = data_changed["Class"].values

print("\nСтандартизированный исходный набор данных:")
display(data_scaled.head())

print("\nСтандартизированный изменённый набор данных:")
display(data_changed_scaled.head())

```

Стандартизированный исходный набор данных:

	0	1	2	3	4	5	6
\							
0	-0.342434	0.330885	0.712859	-0.0469	0.011565	-0.350266	-0.291794
1	0.345359	0.051909	0.435130	-0.0469	-0.256117	0.672399	0.244743
2	-0.145921	-0.165072	0.851723	-0.0469	1.364846	0.343685	0.193644
3	-0.342434	-0.165072	-0.556761	-0.0469	0.472573	-0.350266	0.500237
4	-0.342434	-0.165072	-0.556761	-0.0469	0.472573	-0.350266	0.500237
	7	8	9	...	48	49	50


```

51 \
0 -0.262562 -0.323302 -0.371364 ... -0.158453 -0.514307 -0.155198
0.624007
1 -0.088010 -0.323302 1.086711 ... -0.158453 -0.026007 -0.155198
0.126203
2 0.036670 1.974017 0.016422 ... -0.117376 0.014684 -0.155198
0.008496
3 1.308402 0.789462 0.605857 ... -0.158453 -0.007511 -0.155198 -
0.161934
4 1.308402 0.789462 0.605857 ... -0.158453 -0.014910 -0.155198 -
0.164387

```

	52	53	54	55	56	Class
0	-0.308355	-0.103048	-0.045247	0.045298	-0.008724	1
1	0.423783	0.008763	-0.002443	0.250563	1.228324	1
2	0.440053	-0.079754	0.145921	2.221106	3.258733	1
3	-0.308355	-0.103048	-0.052150	-0.062466	-0.152222	1
4	-0.308355	-0.103048	-0.052150	-0.062466	-0.152222	1

[5 rows x 58 columns]

Стандартизированный изменённый набор данных:

	0	2	4	5	6	7
8 \						
0	-0.342434	0.712859	0.011565	-0.350266	-0.291794	-0.262562 -
	0.323302					
1	0.345359	0.435130	-0.256117	0.672399	0.244743	-0.088010 -
	0.323302					
2	-0.145921	0.851723	1.364846	0.343685	0.193644	0.036670
	1.974017					
3	-0.342434	-0.556761	0.472573	-0.350266	0.500237	1.308402
	0.789462					
4	-0.342434	-0.556761	0.472573	-0.350266	0.500237	1.308402
	0.789462					

	9	10	12	...	41	42	44
45 \							
0	-0.371364	-0.296860	-0.312055	...	-0.1726	-0.205993	-0.297776 -
	0.197387						
1	1.086711	0.745206	1.847391	...	-0.1726	-0.205993	-0.297776 -
	0.197387						
2	0.016422	1.588784	0.086612	...	-0.1726	0.330230	-0.238463 -
	0.131527						
3	0.605857	1.241428	0.717835	...	-0.1726	-0.205993	-0.297776 -
	0.197387						
4	0.605857	1.241428	0.717835	...	-0.1726	-0.205993	-0.297776 -
	0.197387						

	51	52	54	55	56	Class
0	0.624007	-0.308355	-0.045247	0.045298	-0.008724	1
1	0.126203	0.423783	-0.002443	0.250563	1.228324	1
2	0.008496	0.440053	0.145921	2.221106	3.258733	1
3	-0.161934	-0.308355	-0.052150	-0.062466	-0.152222	1
4	-0.164387	-0.308355	-0.052150	-0.062466	-0.152222	1

[5 rows x 42 columns]

Разделение данных на обучающую, валидационную и тестовую выборки для всех наборов данных

```
from sklearn.model_selection import train_test_split
```

```
def split_dataset(df, target='Class', test_size=0.2, val_size=0.2,
random_state=42):
```

"""

Параметры:

- *df: исходный DataFrame.*
- *target: имя столбца с целевой переменной.*
- *test_size: доля выборки для теста.*
- *val_size: доля обучающей выборки, выделенная для валидации.*
- *random_state: фиксированное значение для воспроизводимости.*

Возвращает:

- *train: обучающая выборка.*
- *val: валидационная выборка.*
- *test: тестовая выборка.*

"""

Сначала выделяем тестовую выборку

```
train_val, test = train_test_split(df, test_size=test_size,
random_state=random_state, stratify=df[target])
```

Далее из оставшейся выборки выделяем валидационную

```
train, val = train_test_split(train_val, test_size=val_size,
random_state=random_state, stratify=train_val[target])
```

```
return train, val, test
```

Разбиваем исходный набор данных

```
train_data, val_data, test_data = split_dataset(data, target='Class')
print("data:")
print("Обучающая выборка:", train_data.shape)
print("Валидационная выборка:", val_data.shape)
print("Тестовая выборка:", test_data.shape)
```

Разбиваем стандартизированный исходный набор данных

```
train_data_scaled, val_data_scaled, test_data_scaled =
split_dataset(data_scaled, target='Class')
print("\ndata_scaled:")
```

```

print("Обучающая выборка:", train_data_scaled.shape)
print("Валидационная выборка:", val_data_scaled.shape)
print("Тестовая выборка:", test_data_scaled.shape)

# Разбиваем измененный набор данных
train_data_changed, val_data_changed, test_data_changed =
split_dataset(data_changed, target='Class')
print("\ndata_changed:")
print("Обучающая выборка:", train_data_changed.shape)
print("Валидационная выборка:", val_data_changed.shape)
print("Тестовая выборка:", test_data_changed.shape)

# Разбиваем стандартизированный измененный набор данных
train_data_changed_scaled, val_data_changed_scaled,
test_data_changed_scaled = split_dataset(data_changed_scaled,
target='Class')
print("\ndata_changed_scaled:")
print("Обучающая выборка:", train_data_changed_scaled.shape)
print("Валидационная выборка:", val_data_changed_scaled.shape)
print("Тестовая выборка:", test_data_changed_scaled.shape)

```

data:

```

Обучающая выборка: (2944, 58)
Валидационная выборка: (736, 58)
Тестовая выборка: (921, 58)

```

data_scaled:

```

Обучающая выборка: (2944, 58)
Валидационная выборка: (736, 58)
Тестовая выборка: (921, 58)

```

data_changed:

```

Обучающая выборка: (2944, 42)
Валидационная выборка: (736, 42)
Тестовая выборка: (921, 42)

```

data_changed_scaled:

```

Обучающая выборка: (2944, 42)
Валидационная выборка: (736, 42)
Тестовая выборка: (921, 42)

```

Обучение и оценка моделей (пункт 5)

```

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, Conv1D, Flatten,
Input, Reshape

def build_fully_connected_model(input_shape):
    model = Sequential([

```

```

        Input(shape=input_shape),
        Dense(64, activation='relu'),
        Dropout(0.5),
        Dense(32, activation='relu'),
        Dense(1, activation='sigmoid')
    ])
    model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
    return model

def build_cnn_model(input_dim, filters=32, kernel_size=3,
dense_units=32, dropout_rate=0.5):
    # Для применения Conv1D входные данные нужно преобразовать в форму
(n_features, 1)
    model = Sequential([
        Input(shape=(input_dim,)),
        Reshape((input_dim, 1)),
        Conv1D(filters=filters, kernel_size=kernel_size,
activation='relu'),
        Flatten(),
        Dense(dense_units, activation='relu'),
        Dropout(dropout_rate),
        Dense(1, activation='sigmoid')
    ])
    model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
    return model

# Извлекаем признаки и целевую переменную из исходного набора данных
d_X_train = train_data.drop("Class", axis=1).values
d_Y_train = train_data["Class"].values

# Определяем форму входа для полносвязной модели
input_shape_fc = (d_X_train.shape[1],)

# --- Обучение полносвязной модели (FCNN) ---
d_fc_model = build_fully_connected_model(input_shape_fc)
d_fc_model.fit(d_X_train, d_Y_train, epochs=100, batch_size=32,
verbose=1)

Epoch 1/100
92/92 _____ 0s 314us/step - accuracy: 0.4968 - loss:
8.9939
Epoch 2/100
92/92 _____ 0s 300us/step - accuracy: 0.5728 - loss:
4.7575
Epoch 3/100
92/92 _____ 0s 311us/step - accuracy: 0.6622 - loss:
3.1139

```

```
Epoch 4/100
92/92 _____ 0s 295us/step - accuracy: 0.6993 - loss:
2.2788
Epoch 5/100
92/92 _____ 0s 306us/step - accuracy: 0.7318 - loss:
1.4966
Epoch 6/100
92/92 _____ 0s 296us/step - accuracy: 0.7589 - loss:
1.4519
Epoch 7/100
92/92 _____ 0s 299us/step - accuracy: 0.7546 - loss:
1.1069
Epoch 8/100
92/92 _____ 0s 296us/step - accuracy: 0.7880 - loss:
0.8234
Epoch 9/100
92/92 _____ 0s 289us/step - accuracy: 0.7961 - loss:
0.9060
Epoch 10/100
92/92 _____ 0s 287us/step - accuracy: 0.8073 - loss:
0.7904
Epoch 11/100
92/92 _____ 0s 293us/step - accuracy: 0.8129 - loss:
0.7137
Epoch 12/100
92/92 _____ 0s 282us/step - accuracy: 0.8182 - loss:
0.6685
Epoch 13/100
92/92 _____ 0s 284us/step - accuracy: 0.8149 - loss:
0.5836
Epoch 14/100
92/92 _____ 0s 288us/step - accuracy: 0.8345 - loss:
0.5552
Epoch 15/100
92/92 _____ 0s 282us/step - accuracy: 0.8453 - loss:
0.4743
Epoch 16/100
92/92 _____ 0s 290us/step - accuracy: 0.8499 - loss:
0.5213
Epoch 17/100
92/92 _____ 0s 292us/step - accuracy: 0.8375 - loss:
0.4834
Epoch 18/100
92/92 _____ 0s 293us/step - accuracy: 0.8638 - loss:
0.4116
Epoch 19/100
92/92 _____ 0s 285us/step - accuracy: 0.8555 - loss:
0.4254
Epoch 20/100
```

```
92/92 ————— 0s 302us/step - accuracy: 0.8502 - loss:
0.4288
Epoch 21/100
92/92 ————— 0s 311us/step - accuracy: 0.8609 - loss:
0.4050
Epoch 22/100
92/92 ————— 0s 294us/step - accuracy: 0.8720 - loss:
0.3537
Epoch 23/100
92/92 ————— 0s 288us/step - accuracy: 0.8664 - loss:
0.4084
Epoch 24/100
92/92 ————— 0s 864us/step - accuracy: 0.8747 - loss:
0.3266
Epoch 25/100
92/92 ————— 0s 293us/step - accuracy: 0.8599 - loss:
0.3673
Epoch 26/100
92/92 ————— 0s 292us/step - accuracy: 0.8803 - loss:
0.3286
Epoch 27/100
92/92 ————— 0s 293us/step - accuracy: 0.8928 - loss:
0.2704
Epoch 28/100
92/92 ————— 0s 290us/step - accuracy: 0.8748 - loss:
0.2988
Epoch 29/100
92/92 ————— 0s 303us/step - accuracy: 0.8881 - loss:
0.3050
Epoch 30/100
92/92 ————— 0s 294us/step - accuracy: 0.8819 - loss:
0.3002
Epoch 31/100
92/92 ————— 0s 295us/step - accuracy: 0.9081 - loss:
0.2332
Epoch 32/100
92/92 ————— 0s 292us/step - accuracy: 0.8833 - loss:
0.3017
Epoch 33/100
92/92 ————— 0s 289us/step - accuracy: 0.9055 - loss:
0.2306
Epoch 34/100
92/92 ————— 0s 282us/step - accuracy: 0.8952 - loss:
0.2578
Epoch 35/100
92/92 ————— 0s 290us/step - accuracy: 0.8984 - loss:
0.2644
Epoch 36/100
92/92 ————— 0s 285us/step - accuracy: 0.8804 - loss:
```

```
0.2706
Epoch 37/100
92/92 _____ 0s 282us/step - accuracy: 0.9059 - loss:
0.2380
Epoch 38/100
92/92 _____ 0s 294us/step - accuracy: 0.9095 - loss:
0.2320
Epoch 39/100
92/92 _____ 0s 288us/step - accuracy: 0.9166 - loss:
0.2247
Epoch 40/100
92/92 _____ 0s 292us/step - accuracy: 0.9243 - loss:
0.1965
Epoch 41/100
92/92 _____ 0s 281us/step - accuracy: 0.9034 - loss:
0.2379
Epoch 42/100
92/92 _____ 0s 295us/step - accuracy: 0.9229 - loss:
0.2064
Epoch 43/100
92/92 _____ 0s 303us/step - accuracy: 0.9166 - loss:
0.2361
Epoch 44/100
92/92 _____ 0s 301us/step - accuracy: 0.9126 - loss:
0.2252
Epoch 45/100
92/92 _____ 0s 296us/step - accuracy: 0.9187 - loss:
0.2153
Epoch 46/100
92/92 _____ 0s 299us/step - accuracy: 0.9134 - loss:
0.2152
Epoch 47/100
92/92 _____ 0s 292us/step - accuracy: 0.9164 - loss:
0.2228
Epoch 48/100
92/92 _____ 0s 288us/step - accuracy: 0.9039 - loss:
0.2529
Epoch 49/100
92/92 _____ 0s 294us/step - accuracy: 0.9193 - loss:
0.2112
Epoch 50/100
92/92 _____ 0s 290us/step - accuracy: 0.9234 - loss:
0.2071
Epoch 51/100
92/92 _____ 0s 288us/step - accuracy: 0.9182 - loss:
0.2131
Epoch 52/100
92/92 _____ 0s 286us/step - accuracy: 0.9103 - loss:
0.2070
```

```
Epoch 53/100
92/92 _____ 0s 298us/step - accuracy: 0.9213 - loss:
0.2215
Epoch 54/100
92/92 _____ 0s 300us/step - accuracy: 0.9182 - loss:
0.2188
Epoch 55/100
92/92 _____ 0s 621us/step - accuracy: 0.9095 - loss:
0.2176
Epoch 56/100
92/92 _____ 0s 303us/step - accuracy: 0.9227 - loss:
0.2488
Epoch 57/100
92/92 _____ 0s 289us/step - accuracy: 0.9170 - loss:
0.2221
Epoch 58/100
92/92 _____ 0s 302us/step - accuracy: 0.9124 - loss:
0.2169
Epoch 59/100
92/92 _____ 0s 296us/step - accuracy: 0.9284 - loss:
0.1990
Epoch 60/100
92/92 _____ 0s 293us/step - accuracy: 0.9302 - loss:
0.1857
Epoch 61/100
92/92 _____ 0s 286us/step - accuracy: 0.9287 - loss:
0.1999
Epoch 62/100
92/92 _____ 0s 309us/step - accuracy: 0.9130 - loss:
0.2186
Epoch 63/100
92/92 _____ 0s 284us/step - accuracy: 0.9327 - loss:
0.1847
Epoch 64/100
92/92 _____ 0s 294us/step - accuracy: 0.9174 - loss:
0.2104
Epoch 65/100
92/92 _____ 0s 283us/step - accuracy: 0.9216 - loss:
0.2258
Epoch 66/100
92/92 _____ 0s 298us/step - accuracy: 0.9199 - loss:
0.2157
Epoch 67/100
92/92 _____ 0s 293us/step - accuracy: 0.9298 - loss:
0.1932
Epoch 68/100
92/92 _____ 0s 291us/step - accuracy: 0.9247 - loss:
0.1960
Epoch 69/100
```



```
92/92 ————— 0s 295us/step - accuracy: 0.9241 - loss:
0.2011
Epoch 70/100
92/92 ————— 0s 287us/step - accuracy: 0.9196 - loss:
0.2298
Epoch 71/100
92/92 ————— 0s 288us/step - accuracy: 0.9055 - loss:
0.2492
Epoch 72/100
92/92 ————— 0s 772us/step - accuracy: 0.9278 - loss:
0.2018
Epoch 73/100
92/92 ————— 0s 299us/step - accuracy: 0.9180 - loss:
0.2159
Epoch 74/100
92/92 ————— 0s 290us/step - accuracy: 0.9192 - loss:
0.2056
Epoch 75/100
92/92 ————— 0s 286us/step - accuracy: 0.9158 - loss:
0.2021
Epoch 76/100
92/92 ————— 0s 292us/step - accuracy: 0.9291 - loss:
0.1813
Epoch 77/100
92/92 ————— 0s 289us/step - accuracy: 0.9271 - loss:
0.2048
Epoch 78/100
92/92 ————— 0s 292us/step - accuracy: 0.9336 - loss:
0.1925
Epoch 79/100
92/92 ————— 0s 290us/step - accuracy: 0.9258 - loss:
0.2031
Epoch 80/100
92/92 ————— 0s 285us/step - accuracy: 0.9282 - loss:
0.1916
Epoch 81/100
92/92 ————— 0s 286us/step - accuracy: 0.9154 - loss:
0.2179
Epoch 82/100
92/92 ————— 0s 283us/step - accuracy: 0.9283 - loss:
0.1950
Epoch 83/100
92/92 ————— 0s 292us/step - accuracy: 0.9199 - loss:
0.2010
Epoch 84/100
92/92 ————— 0s 269us/step - accuracy: 0.9283 - loss:
0.1917
Epoch 85/100
92/92 ————— 0s 290us/step - accuracy: 0.9310 - loss:
```

```
0.1790
Epoch 86/100
92/92 _____ 0s 280us/step - accuracy: 0.9301 - loss:
0.1982
Epoch 87/100
92/92 _____ 0s 284us/step - accuracy: 0.9303 - loss:
0.1900
Epoch 88/100
92/92 _____ 0s 293us/step - accuracy: 0.9269 - loss:
0.1977
Epoch 89/100
92/92 _____ 0s 286us/step - accuracy: 0.9385 - loss:
0.1687
Epoch 90/100
92/92 _____ 0s 287us/step - accuracy: 0.9283 - loss:
0.1800
Epoch 91/100
92/92 _____ 0s 449us/step - accuracy: 0.9310 - loss:
0.1797
Epoch 92/100
92/92 _____ 0s 295us/step - accuracy: 0.9340 - loss:
0.1735
Epoch 93/100
92/92 _____ 0s 297us/step - accuracy: 0.9354 - loss:
0.1706
Epoch 94/100
92/92 _____ 0s 278us/step - accuracy: 0.9251 - loss:
0.1946
Epoch 95/100
92/92 _____ 0s 300us/step - accuracy: 0.9365 - loss:
0.1687
Epoch 96/100
92/92 _____ 0s 284us/step - accuracy: 0.9354 - loss:
0.1868
Epoch 97/100
92/92 _____ 0s 302us/step - accuracy: 0.9246 - loss:
0.1989
Epoch 98/100
92/92 _____ 0s 285us/step - accuracy: 0.9342 - loss:
0.1674
Epoch 99/100
92/92 _____ 0s 294us/step - accuracy: 0.9120 - loss:
0.2052
Epoch 100/100
92/92 _____ 0s 416us/step - accuracy: 0.9288 - loss:
0.1714
```

```
<keras.src.callbacks.history.History at 0x358118950>
```

```
# --- Обучение одномерной сверточной модели (1D CNN) ---
```

```
d_cnn_model = build_cnn_model(d_X_train.shape[1])
```

```
d_cnn_model.fit(d_X_train, d_Y_train, epochs=100, batch_size=32,  
verbose=1)
```

```
Epoch 1/100
```

```
92/92 _____ 0s 623us/step - accuracy: 0.6400 - loss:  
1.2339
```

```
Epoch 2/100
```

```
92/92 _____ 0s 686us/step - accuracy: 0.8489 - loss:  
0.3661
```

```
Epoch 3/100
```

```
92/92 _____ 0s 730us/step - accuracy: 0.8806 - loss:  
0.2976
```

```
Epoch 4/100
```

```
92/92 _____ 0s 698us/step - accuracy: 0.8849 - loss:  
0.3084
```

```
Epoch 5/100
```

```
92/92 _____ 0s 696us/step - accuracy: 0.8919 - loss:  
0.2703
```

```
Epoch 6/100
```

```
92/92 _____ 0s 749us/step - accuracy: 0.8725 - loss:  
0.2915
```

```
Epoch 7/100
```

```
92/92 _____ 0s 694us/step - accuracy: 0.8829 - loss:  
0.2764
```

```
Epoch 8/100
```

```
92/92 _____ 0s 725us/step - accuracy: 0.9031 - loss:  
0.2669
```

```
Epoch 9/100
```

```
92/92 _____ 0s 714us/step - accuracy: 0.8955 - loss:  
0.2674
```

```
Epoch 10/100
```

```
92/92 _____ 0s 706us/step - accuracy: 0.9013 - loss:  
0.2490
```

```
Epoch 11/100
```

```
92/92 _____ 0s 741us/step - accuracy: 0.9069 - loss:  
0.2407
```

```
Epoch 12/100
```

```
92/92 _____ 0s 680us/step - accuracy: 0.8907 - loss:  
0.2904
```

```
Epoch 13/100
```

```
92/92 _____ 0s 686us/step - accuracy: 0.8977 - loss:  
0.2460
```

```
Epoch 14/100
```

```
92/92 _____ 0s 729us/step - accuracy: 0.9159 - loss:  
0.2316
```

```
Epoch 15/100
```

```
92/92 _____ 0s 698us/step - accuracy: 0.9139 - loss:  
0.2248
```

```
Epoch 16/100
92/92 _____ 0s 714us/step - accuracy: 0.9123 - loss:
0.2397
Epoch 17/100
92/92 _____ 0s 725us/step - accuracy: 0.9047 - loss:
0.2271
Epoch 18/100
92/92 _____ 0s 704us/step - accuracy: 0.9244 - loss:
0.2260
Epoch 19/100
92/92 _____ 0s 739us/step - accuracy: 0.9032 - loss:
0.2562
Epoch 20/100
92/92 _____ 0s 722us/step - accuracy: 0.9227 - loss:
0.2072
Epoch 21/100
92/92 _____ 0s 699us/step - accuracy: 0.9216 - loss:
0.2058
Epoch 22/100
92/92 _____ 0s 743us/step - accuracy: 0.9191 - loss:
0.2180
Epoch 23/100
92/92 _____ 0s 715us/step - accuracy: 0.9272 - loss:
0.2029
Epoch 24/100
92/92 _____ 0s 712us/step - accuracy: 0.9037 - loss:
0.2554
Epoch 25/100
92/92 _____ 0s 654us/step - accuracy: 0.9189 - loss:
0.2121
Epoch 26/100
92/92 _____ 0s 775us/step - accuracy: 0.9137 - loss:
0.2253
Epoch 27/100
92/92 _____ 0s 708us/step - accuracy: 0.9208 - loss:
0.2125
Epoch 28/100
92/92 _____ 0s 695us/step - accuracy: 0.9181 - loss:
0.2047
Epoch 29/100
92/92 _____ 0s 724us/step - accuracy: 0.9125 - loss:
0.2230
Epoch 30/100
92/92 _____ 0s 716us/step - accuracy: 0.9230 - loss:
0.1999
Epoch 31/100
92/92 _____ 0s 724us/step - accuracy: 0.9191 - loss:
0.1966
Epoch 32/100
```

```
92/92 ————— 0s 708us/step - accuracy: 0.9321 - loss:
0.1775
Epoch 33/100
92/92 ————— 0s 696us/step - accuracy: 0.9325 - loss:
0.1928
Epoch 34/100
92/92 ————— 0s 1ms/step - accuracy: 0.9314 - loss:
0.1822
Epoch 35/100
92/92 ————— 0s 750us/step - accuracy: 0.9376 - loss:
0.1709
Epoch 36/100
92/92 ————— 0s 710us/step - accuracy: 0.9366 - loss:
0.1837
Epoch 37/100
92/92 ————— 0s 738us/step - accuracy: 0.9365 - loss:
0.1907
Epoch 38/100
92/92 ————— 0s 715us/step - accuracy: 0.9337 - loss:
0.1954
Epoch 39/100
92/92 ————— 0s 720us/step - accuracy: 0.9411 - loss:
0.1890
Epoch 40/100
92/92 ————— 0s 718us/step - accuracy: 0.9360 - loss:
0.1806
Epoch 41/100
92/92 ————— 0s 719us/step - accuracy: 0.9360 - loss:
0.1590
Epoch 42/100
92/92 ————— 0s 705us/step - accuracy: 0.9384 - loss:
0.1722
Epoch 43/100
92/92 ————— 0s 708us/step - accuracy: 0.9482 - loss:
0.1766
Epoch 44/100
92/92 ————— 0s 727us/step - accuracy: 0.9349 - loss:
0.1735
Epoch 45/100
92/92 ————— 0s 704us/step - accuracy: 0.9436 - loss:
0.1625
Epoch 46/100
92/92 ————— 0s 704us/step - accuracy: 0.9283 - loss:
0.1973
Epoch 47/100
92/92 ————— 0s 728us/step - accuracy: 0.9433 - loss:
0.1750
Epoch 48/100
92/92 ————— 0s 730us/step - accuracy: 0.9331 - loss:
```

```
0.1832
Epoch 49/100
92/92 _____ 0s 762us/step - accuracy: 0.9297 - loss:
0.1884
Epoch 50/100
92/92 _____ 0s 715us/step - accuracy: 0.9377 - loss:
0.1784
Epoch 51/100
92/92 _____ 0s 730us/step - accuracy: 0.9340 - loss:
0.1703
Epoch 52/100
92/92 _____ 0s 701us/step - accuracy: 0.9411 - loss:
0.1617
Epoch 53/100
92/92 _____ 0s 737us/step - accuracy: 0.9424 - loss:
0.1511
Epoch 54/100
92/92 _____ 0s 721us/step - accuracy: 0.9403 - loss:
0.1456
Epoch 55/100
92/92 _____ 0s 703us/step - accuracy: 0.9419 - loss:
0.1602
Epoch 56/100
92/92 _____ 0s 697us/step - accuracy: 0.9400 - loss:
0.1555
Epoch 57/100
92/92 _____ 0s 716us/step - accuracy: 0.9423 - loss:
0.1546
Epoch 58/100
92/92 _____ 0s 696us/step - accuracy: 0.9368 - loss:
0.1634
Epoch 59/100
92/92 _____ 0s 704us/step - accuracy: 0.9500 - loss:
0.1484
Epoch 60/100
92/92 _____ 0s 696us/step - accuracy: 0.9413 - loss:
0.1516
Epoch 61/100
92/92 _____ 0s 690us/step - accuracy: 0.9447 - loss:
0.1627
Epoch 62/100
92/92 _____ 0s 706us/step - accuracy: 0.9457 - loss:
0.1511
Epoch 63/100
92/92 _____ 0s 709us/step - accuracy: 0.9304 - loss:
0.1806
Epoch 64/100
92/92 _____ 0s 703us/step - accuracy: 0.9378 - loss:
0.1587
```

```
Epoch 65/100
92/92 _____ 0s 702us/step - accuracy: 0.9446 - loss:
0.1584
Epoch 66/100
92/92 _____ 0s 703us/step - accuracy: 0.9490 - loss:
0.1483
Epoch 67/100
92/92 _____ 0s 685us/step - accuracy: 0.9517 - loss:
0.1430
Epoch 68/100
92/92 _____ 0s 861us/step - accuracy: 0.9521 - loss:
0.1381
Epoch 69/100
92/92 _____ 0s 717us/step - accuracy: 0.9522 - loss:
0.1430
Epoch 70/100
92/92 _____ 0s 708us/step - accuracy: 0.9486 - loss:
0.1469
Epoch 71/100
92/92 _____ 0s 705us/step - accuracy: 0.9404 - loss:
0.1511
Epoch 72/100
92/92 _____ 0s 711us/step - accuracy: 0.9466 - loss:
0.1475
Epoch 73/100
92/92 _____ 0s 713us/step - accuracy: 0.9452 - loss:
0.1531
Epoch 74/100
92/92 _____ 0s 709us/step - accuracy: 0.9416 - loss:
0.1546
Epoch 75/100
92/92 _____ 0s 708us/step - accuracy: 0.9474 - loss:
0.1451
Epoch 76/100
92/92 _____ 0s 711us/step - accuracy: 0.9568 - loss:
0.1218
Epoch 77/100
92/92 _____ 0s 712us/step - accuracy: 0.9590 - loss:
0.1356
Epoch 78/100
92/92 _____ 0s 705us/step - accuracy: 0.9590 - loss:
0.1266
Epoch 79/100
92/92 _____ 0s 704us/step - accuracy: 0.9440 - loss:
0.1425
Epoch 80/100
92/92 _____ 0s 704us/step - accuracy: 0.9453 - loss:
0.1474
Epoch 81/100
```

```
92/92 ————— 0s 705us/step - accuracy: 0.9509 - loss:
0.1322
Epoch 82/100
92/92 ————— 0s 714us/step - accuracy: 0.9381 - loss:
0.1843
Epoch 83/100
92/92 ————— 0s 725us/step - accuracy: 0.9461 - loss:
0.1828
Epoch 84/100
92/92 ————— 0s 701us/step - accuracy: 0.9537 - loss:
0.1351
Epoch 85/100
92/92 ————— 0s 705us/step - accuracy: 0.9506 - loss:
0.1263
Epoch 86/100
92/92 ————— 0s 713us/step - accuracy: 0.9436 - loss:
0.1386
Epoch 87/100
92/92 ————— 0s 723us/step - accuracy: 0.9550 - loss:
0.1336
Epoch 88/100
92/92 ————— 0s 716us/step - accuracy: 0.9463 - loss:
0.1294
Epoch 89/100
92/92 ————— 0s 703us/step - accuracy: 0.9440 - loss:
0.1395
Epoch 90/100
92/92 ————— 0s 718us/step - accuracy: 0.9554 - loss:
0.1339
Epoch 91/100
92/92 ————— 0s 1ms/step - accuracy: 0.9587 - loss:
0.1271
Epoch 92/100
92/92 ————— 0s 705us/step - accuracy: 0.9642 - loss:
0.1155
Epoch 93/100
92/92 ————— 0s 679us/step - accuracy: 0.9508 - loss:
0.1398
Epoch 94/100
92/92 ————— 0s 711us/step - accuracy: 0.9484 - loss:
0.1361
Epoch 95/100
92/92 ————— 0s 712us/step - accuracy: 0.9536 - loss:
0.1244
Epoch 96/100
92/92 ————— 0s 700us/step - accuracy: 0.9497 - loss:
0.1413
Epoch 97/100
92/92 ————— 0s 722us/step - accuracy: 0.9529 - loss:
```



```
0.1292
Epoch 98/100
92/92 _____ 0s 711us/step - accuracy: 0.9602 - loss:
0.1229
Epoch 99/100
92/92 _____ 0s 700us/step - accuracy: 0.9551 - loss:
0.1283
Epoch 100/100
92/92 _____ 0s 707us/step - accuracy: 0.9513 - loss:
0.1243
```

```
<keras.src.callbacks.history.History at 0x356444bc0>
```

```
# Извлекаем признаки и целевую переменную из стандартизированного  
исходного набора данных
```

```
sd_X_train = train_data_scaled.drop("Class", axis=1).values  
sd_Y_train = train_data_scaled["Class"].values
```

```
# Определяем форму входа для полносвязной модели  
input_shape_fc = (sd_X_train.shape[1],)
```

```
# --- Обучение полносвязной модели (FCNN) ---
```

```
sd_fc_model = build_fully_connected_model(input_shape_fc)  
sd_fc_model.fit(sd_X_train, sd_Y_train, epochs=100, batch_size=32,  
verbose=1)
```

```
Epoch 1/100  
92/92 _____ 0s 323us/step - accuracy: 0.7142 - loss:  
0.5743  
Epoch 2/100  
92/92 _____ 0s 289us/step - accuracy: 0.8829 - loss:  
0.3336  
Epoch 3/100  
92/92 _____ 0s 288us/step - accuracy: 0.8944 - loss:  
0.2822  
Epoch 4/100  
92/92 _____ 0s 295us/step - accuracy: 0.9176 - loss:  
0.2300  
Epoch 5/100  
92/92 _____ 0s 307us/step - accuracy: 0.9200 - loss:  
0.2217  
Epoch 6/100  
92/92 _____ 0s 299us/step - accuracy: 0.9209 - loss:  
0.2065  
Epoch 7/100  
92/92 _____ 0s 293us/step - accuracy: 0.9251 - loss:  
0.2020  
Epoch 8/100  
92/92 _____ 0s 286us/step - accuracy: 0.9347 - loss:  
0.2011
```

```
Epoch 9/100
92/92 _____ 0s 287us/step - accuracy: 0.9333 - loss:
0.1995
Epoch 10/100
92/92 _____ 0s 280us/step - accuracy: 0.9304 - loss:
0.1880
Epoch 11/100
92/92 _____ 0s 295us/step - accuracy: 0.9368 - loss:
0.1765
Epoch 12/100
92/92 _____ 0s 301us/step - accuracy: 0.9288 - loss:
0.1859
Epoch 13/100
92/92 _____ 0s 294us/step - accuracy: 0.9398 - loss:
0.1644
Epoch 14/100
92/92 _____ 0s 284us/step - accuracy: 0.9334 - loss:
0.1755
Epoch 15/100
92/92 _____ 0s 297us/step - accuracy: 0.9462 - loss:
0.1561
Epoch 16/100
92/92 _____ 0s 282us/step - accuracy: 0.9354 - loss:
0.1825
Epoch 17/100
92/92 _____ 0s 304us/step - accuracy: 0.9385 - loss:
0.1844
Epoch 18/100
92/92 _____ 0s 292us/step - accuracy: 0.9463 - loss:
0.1534
Epoch 19/100
92/92 _____ 0s 293us/step - accuracy: 0.9375 - loss:
0.1807
Epoch 20/100
92/92 _____ 0s 301us/step - accuracy: 0.9466 - loss:
0.1604
Epoch 21/100
92/92 _____ 0s 296us/step - accuracy: 0.9511 - loss:
0.1505
Epoch 22/100
92/92 _____ 0s 301us/step - accuracy: 0.9474 - loss:
0.1525
Epoch 23/100
92/92 _____ 0s 284us/step - accuracy: 0.9364 - loss:
0.1664
Epoch 24/100
92/92 _____ 0s 284us/step - accuracy: 0.9472 - loss:
0.1581
Epoch 25/100
```

```
92/92 ————— 0s 286us/step - accuracy: 0.9529 - loss:
0.1418
Epoch 26/100
92/92 ————— 0s 286us/step - accuracy: 0.9476 - loss:
0.1509
Epoch 27/100
92/92 ————— 0s 300us/step - accuracy: 0.9494 - loss:
0.1424
Epoch 28/100
92/92 ————— 0s 288us/step - accuracy: 0.9391 - loss:
0.1634
Epoch 29/100
92/92 ————— 0s 301us/step - accuracy: 0.9502 - loss:
0.1382
Epoch 30/100
92/92 ————— 0s 293us/step - accuracy: 0.9436 - loss:
0.1541
Epoch 31/100
92/92 ————— 0s 277us/step - accuracy: 0.9524 - loss:
0.1556
Epoch 32/100
92/92 ————— 0s 285us/step - accuracy: 0.9459 - loss:
0.1456
Epoch 33/100
92/92 ————— 0s 284us/step - accuracy: 0.9508 - loss:
0.1376
Epoch 34/100
92/92 ————— 0s 295us/step - accuracy: 0.9506 - loss:
0.1307
Epoch 35/100
92/92 ————— 0s 305us/step - accuracy: 0.9535 - loss:
0.1332
Epoch 36/100
92/92 ————— 0s 286us/step - accuracy: 0.9532 - loss:
0.1371
Epoch 37/100
92/92 ————— 0s 797us/step - accuracy: 0.9495 - loss:
0.1323
Epoch 38/100
92/92 ————— 0s 291us/step - accuracy: 0.9565 - loss:
0.1325
Epoch 39/100
92/92 ————— 0s 286us/step - accuracy: 0.9408 - loss:
0.1484
Epoch 40/100
92/92 ————— 0s 291us/step - accuracy: 0.9518 - loss:
0.1287
Epoch 41/100
92/92 ————— 0s 281us/step - accuracy: 0.9626 - loss:
```

```
0.1227
Epoch 42/100
92/92 _____ 0s 281us/step - accuracy: 0.9551 - loss:
0.1319
Epoch 43/100
92/92 _____ 0s 287us/step - accuracy: 0.9566 - loss:
0.1363
Epoch 44/100
92/92 _____ 0s 293us/step - accuracy: 0.9572 - loss:
0.1245
Epoch 45/100
92/92 _____ 0s 285us/step - accuracy: 0.9578 - loss:
0.1277
Epoch 46/100
92/92 _____ 0s 285us/step - accuracy: 0.9633 - loss:
0.1063
Epoch 47/100
92/92 _____ 0s 296us/step - accuracy: 0.9620 - loss:
0.1060
Epoch 48/100
92/92 _____ 0s 308us/step - accuracy: 0.9554 - loss:
0.1230
Epoch 49/100
92/92 _____ 0s 299us/step - accuracy: 0.9525 - loss:
0.1232
Epoch 50/100
92/92 _____ 0s 296us/step - accuracy: 0.9661 - loss:
0.1060
Epoch 51/100
92/92 _____ 0s 289us/step - accuracy: 0.9514 - loss:
0.1466
Epoch 52/100
92/92 _____ 0s 295us/step - accuracy: 0.9651 - loss:
0.1176
Epoch 53/100
92/92 _____ 0s 279us/step - accuracy: 0.9607 - loss:
0.1070
Epoch 54/100
92/92 _____ 0s 287us/step - accuracy: 0.9690 - loss:
0.1161
Epoch 55/100
92/92 _____ 0s 285us/step - accuracy: 0.9641 - loss:
0.1109
Epoch 56/100
92/92 _____ 0s 298us/step - accuracy: 0.9615 - loss:
0.1132
Epoch 57/100
92/92 _____ 0s 284us/step - accuracy: 0.9663 - loss:
0.1080
```

```
Epoch 58/100
92/92 _____ 0s 277us/step - accuracy: 0.9613 - loss:
0.1144
Epoch 59/100
92/92 _____ 0s 282us/step - accuracy: 0.9588 - loss:
0.1125
Epoch 60/100
92/92 _____ 0s 275us/step - accuracy: 0.9559 - loss:
0.1139
Epoch 61/100
92/92 _____ 0s 298us/step - accuracy: 0.9592 - loss:
0.1161
Epoch 62/100
92/92 _____ 0s 280us/step - accuracy: 0.9615 - loss:
0.1058
Epoch 63/100
92/92 _____ 0s 280us/step - accuracy: 0.9660 - loss:
0.1110
Epoch 64/100
92/92 _____ 0s 286us/step - accuracy: 0.9578 - loss:
0.1158
Epoch 65/100
92/92 _____ 0s 306us/step - accuracy: 0.9667 - loss:
0.0987
Epoch 66/100
92/92 _____ 0s 760us/step - accuracy: 0.9628 - loss:
0.1170
Epoch 67/100
92/92 _____ 0s 321us/step - accuracy: 0.9714 - loss:
0.0909
Epoch 68/100
92/92 _____ 0s 278us/step - accuracy: 0.9691 - loss:
0.1082
Epoch 69/100
92/92 _____ 0s 285us/step - accuracy: 0.9574 - loss:
0.1173
Epoch 70/100
92/92 _____ 0s 279us/step - accuracy: 0.9594 - loss:
0.1049
Epoch 71/100
92/92 _____ 0s 275us/step - accuracy: 0.9663 - loss:
0.1061
Epoch 72/100
92/92 _____ 0s 297us/step - accuracy: 0.9694 - loss:
0.0969
Epoch 73/100
92/92 _____ 0s 326us/step - accuracy: 0.9676 - loss:
0.1062
Epoch 74/100
```

```
92/92 ————— 0s 287us/step - accuracy: 0.9727 - loss:
0.0902
Epoch 75/100
92/92 ————— 0s 294us/step - accuracy: 0.9687 - loss:
0.0952
Epoch 76/100
92/92 ————— 0s 295us/step - accuracy: 0.9695 - loss:
0.0888
Epoch 77/100
92/92 ————— 0s 306us/step - accuracy: 0.9704 - loss:
0.0903
Epoch 78/100
92/92 ————— 0s 312us/step - accuracy: 0.9721 - loss:
0.0829
Epoch 79/100
92/92 ————— 0s 277us/step - accuracy: 0.9724 - loss:
0.0835
Epoch 80/100
92/92 ————— 0s 289us/step - accuracy: 0.9737 - loss:
0.0761
Epoch 81/100
92/92 ————— 0s 282us/step - accuracy: 0.9739 - loss:
0.0816
Epoch 82/100
92/92 ————— 0s 285us/step - accuracy: 0.9716 - loss:
0.0816
Epoch 83/100
92/92 ————— 0s 282us/step - accuracy: 0.9727 - loss:
0.0970
Epoch 84/100
92/92 ————— 0s 267us/step - accuracy: 0.9700 - loss:
0.0891
Epoch 85/100
92/92 ————— 0s 266us/step - accuracy: 0.9651 - loss:
0.0978
Epoch 86/100
92/92 ————— 0s 300us/step - accuracy: 0.9672 - loss:
0.0928
Epoch 87/100
92/92 ————— 0s 296us/step - accuracy: 0.9694 - loss:
0.0988
Epoch 88/100
92/92 ————— 0s 294us/step - accuracy: 0.9613 - loss:
0.1086
Epoch 89/100
92/92 ————— 0s 303us/step - accuracy: 0.9742 - loss:
0.0851
Epoch 90/100
92/92 ————— 0s 289us/step - accuracy: 0.9650 - loss:
```

```

0.0923
Epoch 91/100
92/92 _____ 0s 285us/step - accuracy: 0.9771 - loss:
0.0760
Epoch 92/100
92/92 _____ 0s 280us/step - accuracy: 0.9717 - loss:
0.1001
Epoch 93/100
92/92 _____ 0s 290us/step - accuracy: 0.9783 - loss:
0.0768
Epoch 94/100
92/92 _____ 0s 302us/step - accuracy: 0.9744 - loss:
0.0756
Epoch 95/100
92/92 _____ 0s 283us/step - accuracy: 0.9750 - loss:
0.1024
Epoch 96/100
92/92 _____ 0s 296us/step - accuracy: 0.9708 - loss:
0.0790
Epoch 97/100
92/92 _____ 0s 280us/step - accuracy: 0.9716 - loss:
0.0756
Epoch 98/100
92/92 _____ 0s 290us/step - accuracy: 0.9729 - loss:
0.0904
Epoch 99/100
92/92 _____ 0s 304us/step - accuracy: 0.9725 - loss:
0.0767
Epoch 100/100
92/92 _____ 0s 281us/step - accuracy: 0.9682 - loss:
0.0882

```

```
<keras.src.callbacks.history.History at 0x357f0f1d0>
```

```
# --- Обучение одномерной сверточной модели (1D CNN) ---
```

```
sd_cnn_model = build_cnn_model(sd_X_train.shape[1])
sd_cnn_model.fit(sd_X_train, sd_Y_train, epochs=100, batch_size=32,
verbose=1)
```

```

Epoch 1/100
92/92 _____ 1s 657us/step - accuracy: 0.8250 - loss:
0.4202
Epoch 2/100
92/92 _____ 0s 755us/step - accuracy: 0.9305 - loss:
0.2161
Epoch 3/100
92/92 _____ 0s 756us/step - accuracy: 0.9240 - loss:
0.2142
Epoch 4/100
92/92 _____ 0s 705us/step - accuracy: 0.9345 - loss:

```

```
0.1742
Epoch 5/100
92/92 _____ 0s 695us/step - accuracy: 0.9346 - loss:
0.1896
Epoch 6/100
92/92 _____ 0s 731us/step - accuracy: 0.9318 - loss:
0.1832
Epoch 7/100
92/92 _____ 0s 700us/step - accuracy: 0.9431 - loss:
0.1674
Epoch 8/100
92/92 _____ 0s 702us/step - accuracy: 0.9436 - loss:
0.1683
Epoch 9/100
92/92 _____ 0s 729us/step - accuracy: 0.9440 - loss:
0.1609
Epoch 10/100
92/92 _____ 0s 756us/step - accuracy: 0.9412 - loss:
0.1722
Epoch 11/100
92/92 _____ 0s 753us/step - accuracy: 0.9442 - loss:
0.1468
Epoch 12/100
92/92 _____ 0s 741us/step - accuracy: 0.9474 - loss:
0.1595
Epoch 13/100
92/92 _____ 0s 754us/step - accuracy: 0.9460 - loss:
0.1589
Epoch 14/100
92/92 _____ 0s 808us/step - accuracy: 0.9371 - loss:
0.1741
Epoch 15/100
92/92 _____ 0s 853us/step - accuracy: 0.9488 - loss:
0.1431
Epoch 16/100
92/92 _____ 0s 791us/step - accuracy: 0.9432 - loss:
0.1700
Epoch 17/100
92/92 _____ 0s 798us/step - accuracy: 0.9488 - loss:
0.1498
Epoch 18/100
92/92 _____ 0s 735us/step - accuracy: 0.9421 - loss:
0.1605
Epoch 19/100
92/92 _____ 0s 2ms/step - accuracy: 0.9449 - loss:
0.1425
Epoch 20/100
92/92 _____ 0s 750us/step - accuracy: 0.9499 - loss:
0.1552
```



```
Epoch 21/100
92/92 _____ 0s 709us/step - accuracy: 0.9560 - loss:
0.1277
Epoch 22/100
92/92 _____ 0s 737us/step - accuracy: 0.9517 - loss:
0.1281
Epoch 23/100
92/92 _____ 0s 713us/step - accuracy: 0.9530 - loss:
0.1288
Epoch 24/100
92/92 _____ 0s 729us/step - accuracy: 0.9426 - loss:
0.1397
Epoch 25/100
92/92 _____ 0s 721us/step - accuracy: 0.9470 - loss:
0.1443
Epoch 26/100
92/92 _____ 0s 732us/step - accuracy: 0.9544 - loss:
0.1197
Epoch 27/100
92/92 _____ 0s 715us/step - accuracy: 0.9525 - loss:
0.1449
Epoch 28/100
92/92 _____ 0s 714us/step - accuracy: 0.9519 - loss:
0.1338
Epoch 29/100
92/92 _____ 0s 732us/step - accuracy: 0.9622 - loss:
0.1163
Epoch 30/100
92/92 _____ 0s 715us/step - accuracy: 0.9554 - loss:
0.1259
Epoch 31/100
92/92 _____ 0s 744us/step - accuracy: 0.9552 - loss:
0.1361
Epoch 32/100
92/92 _____ 0s 722us/step - accuracy: 0.9531 - loss:
0.1403
Epoch 33/100
92/92 _____ 0s 723us/step - accuracy: 0.9554 - loss:
0.1324
Epoch 34/100
92/92 _____ 0s 747us/step - accuracy: 0.9569 - loss:
0.1280
Epoch 35/100
92/92 _____ 0s 722us/step - accuracy: 0.9585 - loss:
0.1365
Epoch 36/100
92/92 _____ 0s 725us/step - accuracy: 0.9572 - loss:
0.1171
Epoch 37/100
```

```
92/92 ————— 0s 755us/step - accuracy: 0.9526 - loss:
0.1229
Epoch 38/100
92/92 ————— 0s 721us/step - accuracy: 0.9548 - loss:
0.1259
Epoch 39/100
92/92 ————— 0s 738us/step - accuracy: 0.9588 - loss:
0.1130
Epoch 40/100
92/92 ————— 0s 734us/step - accuracy: 0.9595 - loss:
0.1152
Epoch 41/100
92/92 ————— 0s 747us/step - accuracy: 0.9609 - loss:
0.1084
Epoch 42/100
92/92 ————— 0s 733us/step - accuracy: 0.9596 - loss:
0.1123
Epoch 43/100
92/92 ————— 0s 728us/step - accuracy: 0.9613 - loss:
0.1066
Epoch 44/100
92/92 ————— 0s 725us/step - accuracy: 0.9523 - loss:
0.1238
Epoch 45/100
92/92 ————— 0s 742us/step - accuracy: 0.9591 - loss:
0.1171
Epoch 46/100
92/92 ————— 0s 720us/step - accuracy: 0.9618 - loss:
0.1110
Epoch 47/100
92/92 ————— 0s 743us/step - accuracy: 0.9557 - loss:
0.1159
Epoch 48/100
92/92 ————— 0s 728us/step - accuracy: 0.9640 - loss:
0.1028
Epoch 49/100
92/92 ————— 0s 727us/step - accuracy: 0.9582 - loss:
0.1119
Epoch 50/100
92/92 ————— 0s 702us/step - accuracy: 0.9678 - loss:
0.0944
Epoch 51/100
92/92 ————— 0s 742us/step - accuracy: 0.9620 - loss:
0.1128
Epoch 52/100
92/92 ————— 0s 716us/step - accuracy: 0.9622 - loss:
0.0986
Epoch 53/100
92/92 ————— 0s 741us/step - accuracy: 0.9644 - loss:
```

```
0.1092
Epoch 54/100
92/92 _____ 0s 777us/step - accuracy: 0.9684 - loss:
0.0992
Epoch 55/100
92/92 _____ 0s 791us/step - accuracy: 0.9657 - loss:
0.1022
Epoch 56/100
92/92 _____ 0s 711us/step - accuracy: 0.9662 - loss:
0.1018
Epoch 57/100
92/92 _____ 0s 708us/step - accuracy: 0.9632 - loss:
0.0999
Epoch 58/100
92/92 _____ 0s 738us/step - accuracy: 0.9690 - loss:
0.0812
Epoch 59/100
92/92 _____ 0s 797us/step - accuracy: 0.9648 - loss:
0.1052
Epoch 60/100
92/92 _____ 0s 731us/step - accuracy: 0.9681 - loss:
0.1049
Epoch 61/100
92/92 _____ 0s 709us/step - accuracy: 0.9664 - loss:
0.0914
Epoch 62/100
92/92 _____ 0s 721us/step - accuracy: 0.9691 - loss:
0.0892
Epoch 63/100
92/92 _____ 0s 708us/step - accuracy: 0.9676 - loss:
0.0877
Epoch 64/100
92/92 _____ 0s 759us/step - accuracy: 0.9578 - loss:
0.1048
Epoch 65/100
92/92 _____ 0s 812us/step - accuracy: 0.9703 - loss:
0.0973
Epoch 66/100
92/92 _____ 0s 739us/step - accuracy: 0.9684 - loss:
0.0953
Epoch 67/100
92/92 _____ 0s 847us/step - accuracy: 0.9673 - loss:
0.0914
Epoch 68/100
92/92 _____ 0s 723us/step - accuracy: 0.9744 - loss:
0.0818
Epoch 69/100
92/92 _____ 0s 727us/step - accuracy: 0.9596 - loss:
0.1019
```

```
Epoch 70/100
92/92 _____ 0s 707us/step - accuracy: 0.9700 - loss:
0.0962
Epoch 71/100
92/92 _____ 0s 706us/step - accuracy: 0.9683 - loss:
0.0909
Epoch 72/100
92/92 _____ 0s 712us/step - accuracy: 0.9670 - loss:
0.0893
Epoch 73/100
92/92 _____ 0s 710us/step - accuracy: 0.9673 - loss:
0.0837
Epoch 74/100
92/92 _____ 0s 704us/step - accuracy: 0.9682 - loss:
0.0895
Epoch 75/100
92/92 _____ 0s 708us/step - accuracy: 0.9622 - loss:
0.0907
Epoch 76/100
92/92 _____ 0s 704us/step - accuracy: 0.9707 - loss:
0.0734
Epoch 77/100
92/92 _____ 0s 732us/step - accuracy: 0.9717 - loss:
0.0826
Epoch 78/100
92/92 _____ 0s 699us/step - accuracy: 0.9719 - loss:
0.0832
Epoch 79/100
92/92 _____ 0s 707us/step - accuracy: 0.9716 - loss:
0.0822
Epoch 80/100
92/92 _____ 0s 747us/step - accuracy: 0.9731 - loss:
0.0817
Epoch 81/100
92/92 _____ 0s 707us/step - accuracy: 0.9710 - loss:
0.0834
Epoch 82/100
92/92 _____ 0s 717us/step - accuracy: 0.9670 - loss:
0.0964
Epoch 83/100
92/92 _____ 0s 705us/step - accuracy: 0.9715 - loss:
0.0771
Epoch 84/100
92/92 _____ 0s 728us/step - accuracy: 0.9685 - loss:
0.0783
Epoch 85/100
92/92 _____ 0s 712us/step - accuracy: 0.9682 - loss:
0.0833
Epoch 86/100
```

```
92/92 _____ 0s 727us/step - accuracy: 0.9690 - loss: 0.0989
Epoch 87/100
92/92 _____ 0s 721us/step - accuracy: 0.9747 - loss: 0.0733
Epoch 88/100
92/92 _____ 0s 720us/step - accuracy: 0.9699 - loss: 0.0840
Epoch 89/100
92/92 _____ 0s 718us/step - accuracy: 0.9729 - loss: 0.0754
Epoch 90/100
92/92 _____ 0s 700us/step - accuracy: 0.9717 - loss: 0.0779
Epoch 91/100
92/92 _____ 0s 718us/step - accuracy: 0.9739 - loss: 0.0775
Epoch 92/100
92/92 _____ 0s 1ms/step - accuracy: 0.9689 - loss: 0.0841
Epoch 93/100
92/92 _____ 0s 803us/step - accuracy: 0.9662 - loss: 0.0848
Epoch 94/100
92/92 _____ 0s 685us/step - accuracy: 0.9696 - loss: 0.0794
Epoch 95/100
92/92 _____ 0s 690us/step - accuracy: 0.9760 - loss: 0.0749
Epoch 96/100
92/92 _____ 0s 754us/step - accuracy: 0.9728 - loss: 0.0684
Epoch 97/100
92/92 _____ 0s 727us/step - accuracy: 0.9729 - loss: 0.0718
Epoch 98/100
92/92 _____ 0s 725us/step - accuracy: 0.9706 - loss: 0.0715
Epoch 99/100
92/92 _____ 0s 705us/step - accuracy: 0.9722 - loss: 0.0767
Epoch 100/100
92/92 _____ 0s 739us/step - accuracy: 0.9766 - loss: 0.0641
```

```
<keras.src.callbacks.history.History at 0x359630980>
```

```
# Извлекаем признаки и целевую переменную из изменённого набора данных
dc_X_train = train_data_changed.drop("Class", axis=1).values
dc_Y_train = train_data_changed["Class"].values
```

```
# Определяем форму входа для полносвязной модели
input_shape_fc = (dc_X_train.shape[1],)
```

```
# --- Обучение полносвязной модели (FCNN) ---
```

```
dc_fc_model = build_fully_connected_model(input_shape_fc)
dc_fc_model.fit(dc_X_train, dc_Y_train, epochs=100, batch_size=32,
verbose=1)
```

```
Epoch 1/100
```

```
92/92 _____ 0s 323us/step - accuracy: 0.5094 - loss: 7.5075
```

```
Epoch 2/100
```

```
92/92 _____ 0s 289us/step - accuracy: 0.5653 - loss: 4.0813
```

```
Epoch 3/100
```

```
92/92 _____ 0s 310us/step - accuracy: 0.6660 - loss: 2.6051
```

```
Epoch 4/100
```

```
92/92 _____ 0s 310us/step - accuracy: 0.7013 - loss: 2.4252
```

```
Epoch 5/100
```

```
92/92 _____ 0s 300us/step - accuracy: 0.7058 - loss: 1.8833
```

```
Epoch 6/100
```

```
92/92 _____ 0s 286us/step - accuracy: 0.7326 - loss: 1.3615
```

```
Epoch 7/100
```

```
92/92 _____ 0s 287us/step - accuracy: 0.7528 - loss: 1.3127
```

```
Epoch 8/100
```

```
92/92 _____ 0s 292us/step - accuracy: 0.7490 - loss: 1.2312
```

```
Epoch 9/100
```

```
92/92 _____ 0s 308us/step - accuracy: 0.7716 - loss: 0.8537
```

```
Epoch 10/100
```

```
92/92 _____ 0s 293us/step - accuracy: 0.7795 - loss: 0.8448
```

```
Epoch 11/100
```

```
92/92 _____ 0s 289us/step - accuracy: 0.7826 - loss: 0.7120
```

```
Epoch 12/100
```

```
92/92 _____ 0s 309us/step - accuracy: 0.7882 - loss: 0.7531
```

```
Epoch 13/100
```

```
92/92 _____ 0s 321us/step - accuracy: 0.7959 - loss: 0.7225
```

```
Epoch 14/100
```

```
92/92 _____ 0s 294us/step - accuracy: 0.8118 - loss:
```

```
0.6201
Epoch 15/100
92/92 _____ 0s 300us/step - accuracy: 0.8380 - loss:
0.4835
Epoch 16/100
92/92 _____ 0s 287us/step - accuracy: 0.8304 - loss:
0.5393
Epoch 17/100
92/92 _____ 0s 309us/step - accuracy: 0.8455 - loss:
0.4257
Epoch 18/100
92/92 _____ 0s 288us/step - accuracy: 0.8451 - loss:
0.4503
Epoch 19/100
92/92 _____ 0s 297us/step - accuracy: 0.8657 - loss:
0.3772
Epoch 20/100
92/92 _____ 0s 317us/step - accuracy: 0.8491 - loss:
0.4481
Epoch 21/100
92/92 _____ 0s 294us/step - accuracy: 0.8452 - loss:
0.4621
Epoch 22/100
92/92 _____ 0s 295us/step - accuracy: 0.8542 - loss:
0.3833
Epoch 23/100
92/92 _____ 0s 291us/step - accuracy: 0.8676 - loss:
0.3802
Epoch 24/100
92/92 _____ 0s 286us/step - accuracy: 0.8934 - loss:
0.3289
Epoch 25/100
92/92 _____ 0s 291us/step - accuracy: 0.8770 - loss:
0.3439
Epoch 26/100
92/92 _____ 0s 294us/step - accuracy: 0.8817 - loss:
0.3174
Epoch 27/100
92/92 _____ 0s 299us/step - accuracy: 0.8755 - loss:
0.3214
Epoch 28/100
92/92 _____ 0s 298us/step - accuracy: 0.8900 - loss:
0.3083
Epoch 29/100
92/92 _____ 0s 296us/step - accuracy: 0.8935 - loss:
0.2932
Epoch 30/100
92/92 _____ 0s 316us/step - accuracy: 0.9005 - loss:
0.2870
```

```
Epoch 31/100
92/92 _____ 0s 292us/step - accuracy: 0.8885 - loss:
0.2946
Epoch 32/100
92/92 _____ 0s 305us/step - accuracy: 0.8881 - loss:
0.3309
Epoch 33/100
92/92 _____ 0s 289us/step - accuracy: 0.8859 - loss:
0.3012
Epoch 34/100
92/92 _____ 0s 300us/step - accuracy: 0.9026 - loss:
0.2904
Epoch 35/100
92/92 _____ 0s 292us/step - accuracy: 0.9029 - loss:
0.2723
Epoch 36/100
92/92 _____ 0s 283us/step - accuracy: 0.8908 - loss:
0.3050
Epoch 37/100
92/92 _____ 0s 299us/step - accuracy: 0.9036 - loss:
0.2677
Epoch 38/100
92/92 _____ 0s 307us/step - accuracy: 0.9047 - loss:
0.2874
Epoch 39/100
92/92 _____ 0s 301us/step - accuracy: 0.9062 - loss:
0.2516
Epoch 40/100
92/92 _____ 0s 291us/step - accuracy: 0.9008 - loss:
0.2793
Epoch 41/100
92/92 _____ 0s 294us/step - accuracy: 0.8953 - loss:
0.2585
Epoch 42/100
92/92 _____ 0s 294us/step - accuracy: 0.9110 - loss:
0.2602
Epoch 43/100
92/92 _____ 0s 289us/step - accuracy: 0.9021 - loss:
0.2648
Epoch 44/100
92/92 _____ 0s 285us/step - accuracy: 0.9126 - loss:
0.2395
Epoch 45/100
92/92 _____ 0s 284us/step - accuracy: 0.9118 - loss:
0.2494
Epoch 46/100
92/92 _____ 0s 288us/step - accuracy: 0.8851 - loss:
0.2768
Epoch 47/100
```



```
92/92 ————— 0s 298us/step - accuracy: 0.9001 - loss:
0.2469
Epoch 48/100
92/92 ————— 0s 293us/step - accuracy: 0.9084 - loss:
0.2494
Epoch 49/100
92/92 ————— 0s 299us/step - accuracy: 0.9023 - loss:
0.2545
Epoch 50/100
92/92 ————— 0s 298us/step - accuracy: 0.9109 - loss:
0.2375
Epoch 51/100
92/92 ————— 0s 306us/step - accuracy: 0.9264 - loss:
0.2131
Epoch 52/100
92/92 ————— 0s 299us/step - accuracy: 0.9108 - loss:
0.2347
Epoch 53/100
92/92 ————— 0s 301us/step - accuracy: 0.9119 - loss:
0.2501
Epoch 54/100
92/92 ————— 0s 294us/step - accuracy: 0.9037 - loss:
0.2397
Epoch 55/100
92/92 ————— 0s 285us/step - accuracy: 0.8996 - loss:
0.2786
Epoch 56/100
92/92 ————— 0s 314us/step - accuracy: 0.9058 - loss:
0.2339
Epoch 57/100
92/92 ————— 0s 286us/step - accuracy: 0.9127 - loss:
0.2262
Epoch 58/100
92/92 ————— 0s 281us/step - accuracy: 0.9106 - loss:
0.2426
Epoch 59/100
92/92 ————— 0s 279us/step - accuracy: 0.9199 - loss:
0.2234
Epoch 60/100
92/92 ————— 0s 294us/step - accuracy: 0.9189 - loss:
0.2295
Epoch 61/100
92/92 ————— 0s 301us/step - accuracy: 0.9236 - loss:
0.2044
Epoch 62/100
92/92 ————— 0s 286us/step - accuracy: 0.9103 - loss:
0.2395
Epoch 63/100
92/92 ————— 0s 292us/step - accuracy: 0.9162 - loss:
```

```
0.2249
Epoch 64/100
92/92 _____ 0s 284us/step - accuracy: 0.9150 - loss:
0.2174
Epoch 65/100
92/92 _____ 0s 287us/step - accuracy: 0.9133 - loss:
0.2271
Epoch 66/100
92/92 _____ 0s 282us/step - accuracy: 0.9274 - loss:
0.2181
Epoch 67/100
92/92 _____ 0s 291us/step - accuracy: 0.9301 - loss:
0.1985
Epoch 68/100
92/92 _____ 0s 303us/step - accuracy: 0.9196 - loss:
0.1989
Epoch 69/100
92/92 _____ 0s 290us/step - accuracy: 0.9187 - loss:
0.2148
Epoch 70/100
92/92 _____ 0s 284us/step - accuracy: 0.9231 - loss:
0.2323
Epoch 71/100
92/92 _____ 0s 300us/step - accuracy: 0.9303 - loss:
0.1999
Epoch 72/100
92/92 _____ 0s 285us/step - accuracy: 0.9063 - loss:
0.2337
Epoch 73/100
92/92 _____ 0s 290us/step - accuracy: 0.9248 - loss:
0.2025
Epoch 74/100
92/92 _____ 0s 305us/step - accuracy: 0.9159 - loss:
0.2107
Epoch 75/100
92/92 _____ 0s 281us/step - accuracy: 0.9190 - loss:
0.2112
Epoch 76/100
92/92 _____ 0s 287us/step - accuracy: 0.9254 - loss:
0.2081
Epoch 77/100
92/92 _____ 0s 296us/step - accuracy: 0.9258 - loss:
0.2010
Epoch 78/100
92/92 _____ 0s 287us/step - accuracy: 0.9206 - loss:
0.2199
Epoch 79/100
92/92 _____ 0s 289us/step - accuracy: 0.9223 - loss:
0.2146
```

```
Epoch 80/100
92/92 _____ 0s 287us/step - accuracy: 0.9151 - loss:
0.2259
Epoch 81/100
92/92 _____ 0s 280us/step - accuracy: 0.9179 - loss:
0.2041
Epoch 82/100
92/92 _____ 0s 283us/step - accuracy: 0.9254 - loss:
0.2054
Epoch 83/100
92/92 _____ 0s 290us/step - accuracy: 0.9234 - loss:
0.1986
Epoch 84/100
92/92 _____ 0s 292us/step - accuracy: 0.9151 - loss:
0.2544
Epoch 85/100
92/92 _____ 0s 284us/step - accuracy: 0.9189 - loss:
0.2242
Epoch 86/100
92/92 _____ 0s 285us/step - accuracy: 0.9229 - loss:
0.2021
Epoch 87/100
92/92 _____ 0s 279us/step - accuracy: 0.9215 - loss:
0.1998
Epoch 88/100
92/92 _____ 0s 282us/step - accuracy: 0.9195 - loss:
0.2261
Epoch 89/100
92/92 _____ 0s 299us/step - accuracy: 0.9235 - loss:
0.2025
Epoch 90/100
92/92 _____ 0s 300us/step - accuracy: 0.9184 - loss:
0.2019
Epoch 91/100
92/92 _____ 0s 289us/step - accuracy: 0.9188 - loss:
0.2026
Epoch 92/100
92/92 _____ 0s 284us/step - accuracy: 0.9159 - loss:
0.2140
Epoch 93/100
92/92 _____ 0s 292us/step - accuracy: 0.9072 - loss:
0.2429
Epoch 94/100
92/92 _____ 0s 283us/step - accuracy: 0.9340 - loss:
0.2196
Epoch 95/100
92/92 _____ 0s 303us/step - accuracy: 0.9243 - loss:
0.1966
Epoch 96/100
```

```
92/92 _____ 0s 293us/step - accuracy: 0.9219 - loss: 0.1981
Epoch 97/100
92/92 _____ 0s 290us/step - accuracy: 0.9279 - loss: 0.1836
Epoch 98/100
92/92 _____ 0s 276us/step - accuracy: 0.9348 - loss: 0.1897
Epoch 99/100
92/92 _____ 0s 286us/step - accuracy: 0.9266 - loss: 0.1955
Epoch 100/100
92/92 _____ 0s 293us/step - accuracy: 0.9185 - loss: 0.2106
```

```
<keras.src.callbacks.history.History at 0x359624680>
```

```
# --- Обучение одномерной сверточной модели (1D CNN) ---
```

```
dc_cnn_model = build_cnn_model(dc_X_train.shape[1])
dc_cnn_model.fit(dc_X_train, dc_Y_train, epochs=100, batch_size=32,
verbose=1)
```

```
Epoch 1/100
92/92 _____ 0s 545us/step - accuracy: 0.6524 - loss: 1.2132
Epoch 2/100
92/92 _____ 0s 643us/step - accuracy: 0.8114 - loss: 0.4458
Epoch 3/100
92/92 _____ 0s 632us/step - accuracy: 0.8414 - loss: 0.3654
Epoch 4/100
92/92 _____ 0s 625us/step - accuracy: 0.8733 - loss: 0.3263
Epoch 5/100
92/92 _____ 0s 585us/step - accuracy: 0.8697 - loss: 0.3165
Epoch 6/100
92/92 _____ 0s 572us/step - accuracy: 0.8677 - loss: 0.3188
Epoch 7/100
92/92 _____ 0s 594us/step - accuracy: 0.8856 - loss: 0.2897
Epoch 8/100
92/92 _____ 0s 597us/step - accuracy: 0.8919 - loss: 0.2719
Epoch 9/100
92/92 _____ 0s 591us/step - accuracy: 0.8736 - loss: 0.3013
Epoch 10/100
```

```
92/92 _____ 0s 645us/step - accuracy: 0.9040 - loss:
0.2697
Epoch 11/100
92/92 _____ 0s 616us/step - accuracy: 0.8945 - loss:
0.2678
Epoch 12/100
92/92 _____ 0s 586us/step - accuracy: 0.8912 - loss:
0.2700
Epoch 13/100
92/92 _____ 0s 550us/step - accuracy: 0.9007 - loss:
0.2735
Epoch 14/100
92/92 _____ 0s 577us/step - accuracy: 0.9002 - loss:
0.2537
Epoch 15/100
92/92 _____ 0s 601us/step - accuracy: 0.9114 - loss:
0.2532
Epoch 16/100
92/92 _____ 0s 583us/step - accuracy: 0.9132 - loss:
0.2503
Epoch 17/100
92/92 _____ 0s 623us/step - accuracy: 0.9024 - loss:
0.2599
Epoch 18/100
92/92 _____ 0s 621us/step - accuracy: 0.9051 - loss:
0.2515
Epoch 19/100
92/92 _____ 0s 592us/step - accuracy: 0.9105 - loss:
0.2372
Epoch 20/100
92/92 _____ 0s 570us/step - accuracy: 0.9282 - loss:
0.2139
Epoch 21/100
92/92 _____ 0s 647us/step - accuracy: 0.9144 - loss:
0.2258
Epoch 22/100
92/92 _____ 0s 629us/step - accuracy: 0.9075 - loss:
0.2488
Epoch 23/100
92/92 _____ 0s 1ms/step - accuracy: 0.9045 - loss:
0.3277
Epoch 24/100
92/92 _____ 0s 632us/step - accuracy: 0.8880 - loss:
0.2761
Epoch 25/100
92/92 _____ 0s 617us/step - accuracy: 0.8989 - loss:
0.2626
Epoch 26/100
92/92 _____ 0s 616us/step - accuracy: 0.9083 - loss:
```

```
0.2521
Epoch 27/100
92/92 _____ 0s 626us/step - accuracy: 0.9155 - loss:
0.2211
Epoch 28/100
92/92 _____ 0s 610us/step - accuracy: 0.9200 - loss:
0.2262
Epoch 29/100
92/92 _____ 0s 625us/step - accuracy: 0.9169 - loss:
0.2179
Epoch 30/100
92/92 _____ 0s 577us/step - accuracy: 0.9060 - loss:
0.2482
Epoch 31/100
92/92 _____ 0s 583us/step - accuracy: 0.9156 - loss:
0.2204
Epoch 32/100
92/92 _____ 0s 626us/step - accuracy: 0.9203 - loss:
0.2163
Epoch 33/100
92/92 _____ 0s 606us/step - accuracy: 0.9186 - loss:
0.2309
Epoch 34/100
92/92 _____ 0s 601us/step - accuracy: 0.9235 - loss:
0.2408
Epoch 35/100
92/92 _____ 0s 599us/step - accuracy: 0.9263 - loss:
0.2043
Epoch 36/100
92/92 _____ 0s 602us/step - accuracy: 0.9257 - loss:
0.1947
Epoch 37/100
92/92 _____ 0s 602us/step - accuracy: 0.9110 - loss:
0.2335
Epoch 38/100
92/92 _____ 0s 612us/step - accuracy: 0.9217 - loss:
0.2204
Epoch 39/100
92/92 _____ 0s 610us/step - accuracy: 0.9282 - loss:
0.2185
Epoch 40/100
92/92 _____ 0s 577us/step - accuracy: 0.9137 - loss:
0.2223
Epoch 41/100
92/92 _____ 0s 645us/step - accuracy: 0.9266 - loss:
0.2053
Epoch 42/100
92/92 _____ 0s 607us/step - accuracy: 0.9239 - loss:
0.1995
```

```
Epoch 43/100
92/92 _____ 0s 595us/step - accuracy: 0.9191 - loss:
0.2237
Epoch 44/100
92/92 _____ 0s 602us/step - accuracy: 0.9211 - loss:
0.2154
Epoch 45/100
92/92 _____ 0s 619us/step - accuracy: 0.9218 - loss:
0.1981
Epoch 46/100
92/92 _____ 0s 585us/step - accuracy: 0.9126 - loss:
0.2257
Epoch 47/100
92/92 _____ 0s 619us/step - accuracy: 0.9189 - loss:
0.1957
Epoch 48/100
92/92 _____ 0s 619us/step - accuracy: 0.9259 - loss:
0.1918
Epoch 49/100
92/92 _____ 0s 601us/step - accuracy: 0.9252 - loss:
0.2206
Epoch 50/100
92/92 _____ 0s 607us/step - accuracy: 0.9314 - loss:
0.1890
Epoch 51/100
92/92 _____ 0s 595us/step - accuracy: 0.9263 - loss:
0.2634
Epoch 52/100
92/92 _____ 0s 580us/step - accuracy: 0.9178 - loss:
0.1992
Epoch 53/100
92/92 _____ 0s 658us/step - accuracy: 0.9215 - loss:
0.1949
Epoch 54/100
92/92 _____ 0s 658us/step - accuracy: 0.9254 - loss:
0.1951
Epoch 55/100
92/92 _____ 0s 633us/step - accuracy: 0.9424 - loss:
0.1701
Epoch 56/100
92/92 _____ 0s 625us/step - accuracy: 0.9218 - loss:
0.2023
Epoch 57/100
92/92 _____ 0s 623us/step - accuracy: 0.9375 - loss:
0.1669
Epoch 58/100
92/92 _____ 0s 659us/step - accuracy: 0.9401 - loss:
0.1634
Epoch 59/100
```

```
92/92 _____ 0s 715us/step - accuracy: 0.9507 - loss:
0.1578
Epoch 60/100
92/92 _____ 0s 658us/step - accuracy: 0.9311 - loss:
0.1931
Epoch 61/100
92/92 _____ 0s 709us/step - accuracy: 0.9391 - loss:
0.1868
Epoch 62/100
92/92 _____ 0s 620us/step - accuracy: 0.9395 - loss:
0.1663
Epoch 63/100
92/92 _____ 0s 615us/step - accuracy: 0.9380 - loss:
0.1699
Epoch 64/100
92/92 _____ 0s 687us/step - accuracy: 0.9383 - loss:
0.1754
Epoch 65/100
92/92 _____ 0s 585us/step - accuracy: 0.9460 - loss:
0.1677
Epoch 66/100
92/92 _____ 0s 577us/step - accuracy: 0.9454 - loss:
0.1656
Epoch 67/100
92/92 _____ 0s 604us/step - accuracy: 0.9499 - loss:
0.1582
Epoch 68/100
92/92 _____ 0s 610us/step - accuracy: 0.9436 - loss:
0.1700
Epoch 69/100
92/92 _____ 0s 605us/step - accuracy: 0.9424 - loss:
0.1733
Epoch 70/100
92/92 _____ 0s 592us/step - accuracy: 0.9406 - loss:
0.1662
Epoch 71/100
92/92 _____ 0s 589us/step - accuracy: 0.9478 - loss:
0.1564
Epoch 72/100
92/92 _____ 0s 608us/step - accuracy: 0.9356 - loss:
0.2094
Epoch 73/100
92/92 _____ 0s 610us/step - accuracy: 0.9380 - loss:
0.1723
Epoch 74/100
92/92 _____ 0s 587us/step - accuracy: 0.9406 - loss:
0.1902
Epoch 75/100
92/92 _____ 0s 575us/step - accuracy: 0.9371 - loss:
```



```
0.1831
Epoch 76/100
92/92 _____ 0s 600us/step - accuracy: 0.9458 - loss:
0.1535
Epoch 77/100
92/92 _____ 0s 598us/step - accuracy: 0.9327 - loss:
0.1706
Epoch 78/100
92/92 _____ 0s 576us/step - accuracy: 0.9463 - loss:
0.1559
Epoch 79/100
92/92 _____ 0s 611us/step - accuracy: 0.9451 - loss:
0.1534
Epoch 80/100
92/92 _____ 0s 604us/step - accuracy: 0.9399 - loss:
0.1616
Epoch 81/100
92/92 _____ 0s 614us/step - accuracy: 0.9457 - loss:
0.1570
Epoch 82/100
92/92 _____ 0s 614us/step - accuracy: 0.9488 - loss:
0.1538
Epoch 83/100
92/92 _____ 0s 620us/step - accuracy: 0.9495 - loss:
0.1566
Epoch 84/100
92/92 _____ 0s 597us/step - accuracy: 0.9482 - loss:
0.1350
Epoch 85/100
92/92 _____ 0s 579us/step - accuracy: 0.9466 - loss:
0.1620
Epoch 86/100
92/92 _____ 0s 574us/step - accuracy: 0.9467 - loss:
0.1593
Epoch 87/100
92/92 _____ 0s 603us/step - accuracy: 0.9521 - loss:
0.1494
Epoch 88/100
92/92 _____ 0s 589us/step - accuracy: 0.9499 - loss:
0.1474
Epoch 89/100
92/92 _____ 0s 614us/step - accuracy: 0.9578 - loss:
0.1522
Epoch 90/100
92/92 _____ 0s 617us/step - accuracy: 0.9502 - loss:
0.1410
Epoch 91/100
92/92 _____ 0s 592us/step - accuracy: 0.9485 - loss:
0.1450
```

```
Epoch 92/100
92/92 _____ 0s 1ms/step - accuracy: 0.9483 - loss:
0.1483
Epoch 93/100
92/92 _____ 0s 608us/step - accuracy: 0.9518 - loss:
0.1523
Epoch 94/100
92/92 _____ 0s 599us/step - accuracy: 0.9487 - loss:
0.1524
Epoch 95/100
92/92 _____ 0s 595us/step - accuracy: 0.9444 - loss:
0.1591
Epoch 96/100
92/92 _____ 0s 575us/step - accuracy: 0.9510 - loss:
0.1406
Epoch 97/100
92/92 _____ 0s 612us/step - accuracy: 0.9467 - loss:
0.1432
Epoch 98/100
92/92 _____ 0s 611us/step - accuracy: 0.9413 - loss:
0.1580
Epoch 99/100
92/92 _____ 0s 581us/step - accuracy: 0.9388 - loss:
0.1714
Epoch 100/100
92/92 _____ 0s 599us/step - accuracy: 0.9426 - loss:
0.1520
```

```
<keras.src.callbacks.history.History at 0x31c3cc680>
```

```
# Извлекаем признаки и целевую переменную из стандартизированного
изменённого набора данных
```

```
sdc_X_train = train_data_changed_scaled.drop("Class", axis=1).values
sdc_Y_train = train_data_changed_scaled["Class"].values
```

```
# Определяем форму входа для полносвязной модели
```

```
input_shape_fc = (sdc_X_train.shape[1],)
```

```
# --- Обучение полносвязной модели (FCNN) ---
```

```
sdc_fc_model = build_fully_connected_model(input_shape_fc)
sdc_fc_model.fit(sdc_X_train, sdc_Y_train, epochs=100, batch_size=32,
verbose=1)
```

```
Epoch 1/100
92/92 _____ 0s 291us/step - accuracy: 0.7122 - loss:
0.6004
Epoch 2/100
92/92 _____ 0s 276us/step - accuracy: 0.8695 - loss:
0.3399
Epoch 3/100
```

```
92/92 _____ 0s 291us/step - accuracy: 0.8937 - loss:
0.2821
Epoch 4/100
92/92 _____ 0s 304us/step - accuracy: 0.9062 - loss:
0.2524
Epoch 5/100
92/92 _____ 0s 301us/step - accuracy: 0.9129 - loss:
0.2412
Epoch 6/100
92/92 _____ 0s 293us/step - accuracy: 0.9260 - loss:
0.2171
Epoch 7/100
92/92 _____ 0s 285us/step - accuracy: 0.9167 - loss:
0.2087
Epoch 8/100
92/92 _____ 0s 290us/step - accuracy: 0.9224 - loss:
0.2036
Epoch 9/100
92/92 _____ 0s 290us/step - accuracy: 0.9301 - loss:
0.2017
Epoch 10/100
92/92 _____ 0s 298us/step - accuracy: 0.9131 - loss:
0.2345
Epoch 11/100
92/92 _____ 0s 284us/step - accuracy: 0.9174 - loss:
0.2197
Epoch 12/100
92/92 _____ 0s 294us/step - accuracy: 0.9337 - loss:
0.1830
Epoch 13/100
92/92 _____ 0s 283us/step - accuracy: 0.9344 - loss:
0.1920
Epoch 14/100
92/92 _____ 0s 283us/step - accuracy: 0.9315 - loss:
0.2086
Epoch 15/100
92/92 _____ 0s 298us/step - accuracy: 0.9414 - loss:
0.1720
Epoch 16/100
92/92 _____ 0s 284us/step - accuracy: 0.9265 - loss:
0.1891
Epoch 17/100
92/92 _____ 0s 284us/step - accuracy: 0.9393 - loss:
0.1753
Epoch 18/100
92/92 _____ 0s 281us/step - accuracy: 0.9275 - loss:
0.2109
Epoch 19/100
92/92 _____ 0s 293us/step - accuracy: 0.9271 - loss:
```

```
0.1983
Epoch 20/100
92/92 _____ 0s 286us/step - accuracy: 0.9363 - loss:
0.1857
Epoch 21/100
92/92 _____ 0s 293us/step - accuracy: 0.9361 - loss:
0.1829
Epoch 22/100
92/92 _____ 0s 295us/step - accuracy: 0.9448 - loss:
0.1552
Epoch 23/100
92/92 _____ 0s 301us/step - accuracy: 0.9260 - loss:
0.1990
Epoch 24/100
92/92 _____ 0s 289us/step - accuracy: 0.9399 - loss:
0.1686
Epoch 25/100
92/92 _____ 0s 288us/step - accuracy: 0.9420 - loss:
0.1734
Epoch 26/100
92/92 _____ 0s 298us/step - accuracy: 0.9390 - loss:
0.1642
Epoch 27/100
92/92 _____ 0s 296us/step - accuracy: 0.9392 - loss:
0.1614
Epoch 28/100
92/92 _____ 0s 282us/step - accuracy: 0.9416 - loss:
0.1621
Epoch 29/100
92/92 _____ 0s 301us/step - accuracy: 0.9474 - loss:
0.1665
Epoch 30/100
92/92 _____ 0s 286us/step - accuracy: 0.9445 - loss:
0.1491
Epoch 31/100
92/92 _____ 0s 298us/step - accuracy: 0.9488 - loss:
0.1648
Epoch 32/100
92/92 _____ 0s 297us/step - accuracy: 0.9499 - loss:
0.1456
Epoch 33/100
92/92 _____ 0s 288us/step - accuracy: 0.9504 - loss:
0.1477
Epoch 34/100
92/92 _____ 0s 288us/step - accuracy: 0.9432 - loss:
0.1540
Epoch 35/100
92/92 _____ 0s 291us/step - accuracy: 0.9489 - loss:
0.1504
```

```
Epoch 36/100
92/92 _____ 0s 293us/step - accuracy: 0.9531 - loss:
0.1453
Epoch 37/100
92/92 _____ 0s 285us/step - accuracy: 0.9457 - loss:
0.1501
Epoch 38/100
92/92 _____ 0s 296us/step - accuracy: 0.9541 - loss:
0.1312
Epoch 39/100
92/92 _____ 0s 288us/step - accuracy: 0.9501 - loss:
0.1437
Epoch 40/100
92/92 _____ 0s 307us/step - accuracy: 0.9501 - loss:
0.1479
Epoch 41/100
92/92 _____ 0s 287us/step - accuracy: 0.9532 - loss:
0.1349
Epoch 42/100
92/92 _____ 0s 287us/step - accuracy: 0.9479 - loss:
0.1447
Epoch 43/100
92/92 _____ 0s 304us/step - accuracy: 0.9575 - loss:
0.1297
Epoch 44/100
92/92 _____ 0s 284us/step - accuracy: 0.9489 - loss:
0.1451
Epoch 45/100
92/92 _____ 0s 295us/step - accuracy: 0.9495 - loss:
0.1402
Epoch 46/100
92/92 _____ 0s 298us/step - accuracy: 0.9530 - loss:
0.1282
Epoch 47/100
92/92 _____ 0s 302us/step - accuracy: 0.9503 - loss:
0.1387
Epoch 48/100
92/92 _____ 0s 288us/step - accuracy: 0.9512 - loss:
0.1469
Epoch 49/100
92/92 _____ 0s 290us/step - accuracy: 0.9378 - loss:
0.1561
Epoch 50/100
92/92 _____ 0s 285us/step - accuracy: 0.9572 - loss:
0.1205
Epoch 51/100
92/92 _____ 0s 289us/step - accuracy: 0.9515 - loss:
0.1420
Epoch 52/100
```

```
92/92 ————— 0s 288us/step - accuracy: 0.9583 - loss:
0.1224
Epoch 53/100
92/92 ————— 0s 277us/step - accuracy: 0.9583 - loss:
0.1293
Epoch 54/100
92/92 ————— 0s 288us/step - accuracy: 0.9560 - loss:
0.1231
Epoch 55/100
92/92 ————— 0s 276us/step - accuracy: 0.9564 - loss:
0.1250
Epoch 56/100
92/92 ————— 0s 285us/step - accuracy: 0.9581 - loss:
0.1242
Epoch 57/100
92/92 ————— 0s 287us/step - accuracy: 0.9621 - loss:
0.1198
Epoch 58/100
92/92 ————— 0s 294us/step - accuracy: 0.9544 - loss:
0.1462
Epoch 59/100
92/92 ————— 0s 279us/step - accuracy: 0.9532 - loss:
0.1243
Epoch 60/100
92/92 ————— 0s 277us/step - accuracy: 0.9556 - loss:
0.1248
Epoch 61/100
92/92 ————— 0s 285us/step - accuracy: 0.9614 - loss:
0.1160
Epoch 62/100
92/92 ————— 0s 287us/step - accuracy: 0.9597 - loss:
0.1159
Epoch 63/100
92/92 ————— 0s 279us/step - accuracy: 0.9632 - loss:
0.1356
Epoch 64/100
92/92 ————— 0s 293us/step - accuracy: 0.9541 - loss:
0.1208
Epoch 65/100
92/92 ————— 0s 286us/step - accuracy: 0.9589 - loss:
0.1230
Epoch 66/100
92/92 ————— 0s 289us/step - accuracy: 0.9624 - loss:
0.1051
Epoch 67/100
92/92 ————— 0s 287us/step - accuracy: 0.9617 - loss:
0.1116
Epoch 68/100
92/92 ————— 0s 279us/step - accuracy: 0.9498 - loss:
```

```
0.1317
Epoch 69/100
92/92 _____ 0s 288us/step - accuracy: 0.9571 - loss:
0.1178
Epoch 70/100
92/92 _____ 0s 281us/step - accuracy: 0.9536 - loss:
0.1326
Epoch 71/100
92/92 _____ 0s 283us/step - accuracy: 0.9592 - loss:
0.1042
Epoch 72/100
92/92 _____ 0s 289us/step - accuracy: 0.9588 - loss:
0.1144
Epoch 73/100
92/92 _____ 0s 285us/step - accuracy: 0.9590 - loss:
0.1193
Epoch 74/100
92/92 _____ 0s 292us/step - accuracy: 0.9573 - loss:
0.1162
Epoch 75/100
92/92 _____ 0s 282us/step - accuracy: 0.9614 - loss:
0.1156
Epoch 76/100
92/92 _____ 0s 278us/step - accuracy: 0.9573 - loss:
0.1263
Epoch 77/100
92/92 _____ 0s 280us/step - accuracy: 0.9573 - loss:
0.1138
Epoch 78/100
92/92 _____ 0s 303us/step - accuracy: 0.9603 - loss:
0.1045
Epoch 79/100
92/92 _____ 0s 279us/step - accuracy: 0.9591 - loss:
0.1152
Epoch 80/100
92/92 _____ 0s 281us/step - accuracy: 0.9614 - loss:
0.1211
Epoch 81/100
92/92 _____ 0s 276us/step - accuracy: 0.9611 - loss:
0.1226
Epoch 82/100
92/92 _____ 0s 287us/step - accuracy: 0.9606 - loss:
0.0993
Epoch 83/100
92/92 _____ 0s 283us/step - accuracy: 0.9579 - loss:
0.1162
Epoch 84/100
92/92 _____ 0s 276us/step - accuracy: 0.9638 - loss:
0.1073
```

```
Epoch 85/100
92/92 _____ 0s 292us/step - accuracy: 0.9637 - loss:
0.1049
Epoch 86/100
92/92 _____ 0s 284us/step - accuracy: 0.9669 - loss:
0.1010
Epoch 87/100
92/92 _____ 0s 288us/step - accuracy: 0.9565 - loss:
0.1222
Epoch 88/100
92/92 _____ 0s 287us/step - accuracy: 0.9580 - loss:
0.1201
Epoch 89/100
92/92 _____ 0s 301us/step - accuracy: 0.9595 - loss:
0.1120
Epoch 90/100
92/92 _____ 0s 293us/step - accuracy: 0.9609 - loss:
0.1047
Epoch 91/100
92/92 _____ 0s 283us/step - accuracy: 0.9554 - loss:
0.1044
Epoch 92/100
92/92 _____ 0s 284us/step - accuracy: 0.9684 - loss:
0.0954
Epoch 93/100
92/92 _____ 0s 288us/step - accuracy: 0.9623 - loss:
0.1023
Epoch 94/100
92/92 _____ 0s 292us/step - accuracy: 0.9576 - loss:
0.1079
Epoch 95/100
92/92 _____ 0s 284us/step - accuracy: 0.9580 - loss:
0.1088
Epoch 96/100
92/92 _____ 0s 284us/step - accuracy: 0.9599 - loss:
0.1216
Epoch 97/100
92/92 _____ 0s 288us/step - accuracy: 0.9631 - loss:
0.1043
Epoch 98/100
92/92 _____ 0s 291us/step - accuracy: 0.9583 - loss:
0.1090
Epoch 99/100
92/92 _____ 0s 303us/step - accuracy: 0.9667 - loss:
0.0950
Epoch 100/100
92/92 _____ 0s 285us/step - accuracy: 0.9658 - loss:
0.1151
```

```
<keras.src.callbacks.history.History at 0x32bd6a1e0>
```



```
# --- Обучение одномерной сверточной модели (1D CNN) ---  
sdc_cnn_model = build_cnn_model(sdc_X_train.shape[1])  
sdc_cnn_model.fit(sdc_X_train, sdc_Y_train, epochs=100, batch_size=32,  
verbose=1)
```

Epoch 1/100

92/92 _____ 0s 578us/step - accuracy: 0.8414 - loss:
0.4198

Epoch 2/100

92/92 _____ 0s 598us/step - accuracy: 0.9280 - loss:
0.2309

Epoch 3/100

92/92 _____ 0s 582us/step - accuracy: 0.9167 - loss:
0.2353

Epoch 4/100

92/92 _____ 0s 636us/step - accuracy: 0.9259 - loss:
0.2089

Epoch 5/100

92/92 _____ 0s 617us/step - accuracy: 0.9329 - loss:
0.1859

Epoch 6/100

92/92 _____ 0s 614us/step - accuracy: 0.9387 - loss:
0.1779

Epoch 7/100

92/92 _____ 0s 629us/step - accuracy: 0.9364 - loss:
0.1820

Epoch 8/100

92/92 _____ 0s 577us/step - accuracy: 0.9443 - loss:
0.1634

Epoch 9/100

92/92 _____ 0s 642us/step - accuracy: 0.9398 - loss:
0.1761

Epoch 10/100

92/92 _____ 0s 634us/step - accuracy: 0.9389 - loss:
0.1592

Epoch 11/100

92/92 _____ 0s 620us/step - accuracy: 0.9414 - loss:
0.1647

Epoch 12/100

92/92 _____ 0s 626us/step - accuracy: 0.9387 - loss:
0.1684

Epoch 13/100

92/92 _____ 0s 585us/step - accuracy: 0.9426 - loss:
0.1687

Epoch 14/100

92/92 _____ 0s 577us/step - accuracy: 0.9423 - loss:
0.1545

Epoch 15/100

92/92 _____ 0s 597us/step - accuracy: 0.9420 - loss:
0.1628

```
Epoch 16/100
92/92 _____ 0s 1ms/step - accuracy: 0.9463 - loss:
0.1442
Epoch 17/100
92/92 _____ 0s 641us/step - accuracy: 0.9458 - loss:
0.1422
Epoch 18/100
92/92 _____ 0s 619us/step - accuracy: 0.9428 - loss:
0.1586
Epoch 19/100
92/92 _____ 0s 597us/step - accuracy: 0.9320 - loss:
0.1692
Epoch 20/100
92/92 _____ 0s 618us/step - accuracy: 0.9446 - loss:
0.1555
Epoch 21/100
92/92 _____ 0s 603us/step - accuracy: 0.9467 - loss:
0.1479
Epoch 22/100
92/92 _____ 0s 608us/step - accuracy: 0.9417 - loss:
0.1505
Epoch 23/100
92/92 _____ 0s 595us/step - accuracy: 0.9475 - loss:
0.1373
Epoch 24/100
92/92 _____ 0s 612us/step - accuracy: 0.9479 - loss:
0.1449
Epoch 25/100
92/92 _____ 0s 620us/step - accuracy: 0.9473 - loss:
0.1473
Epoch 26/100
92/92 _____ 0s 606us/step - accuracy: 0.9468 - loss:
0.1361
Epoch 27/100
92/92 _____ 0s 590us/step - accuracy: 0.9496 - loss:
0.1293
Epoch 28/100
92/92 _____ 0s 627us/step - accuracy: 0.9467 - loss:
0.1462
Epoch 29/100
92/92 _____ 0s 575us/step - accuracy: 0.9433 - loss:
0.1440
Epoch 30/100
92/92 _____ 0s 702us/step - accuracy: 0.9472 - loss:
0.1567
Epoch 31/100
92/92 _____ 0s 608us/step - accuracy: 0.9460 - loss:
0.1476
Epoch 32/100
```

```
92/92 ————— 0s 622us/step - accuracy: 0.9503 - loss:
0.1394
Epoch 33/100
92/92 ————— 0s 651us/step - accuracy: 0.9459 - loss:
0.1370
Epoch 34/100
92/92 ————— 0s 627us/step - accuracy: 0.9494 - loss:
0.1453
Epoch 35/100
92/92 ————— 0s 621us/step - accuracy: 0.9453 - loss:
0.1445
Epoch 36/100
92/92 ————— 0s 627us/step - accuracy: 0.9546 - loss:
0.1375
Epoch 37/100
92/92 ————— 0s 613us/step - accuracy: 0.9578 - loss:
0.1309
Epoch 38/100
92/92 ————— 0s 635us/step - accuracy: 0.9545 - loss:
0.1362
Epoch 39/100
92/92 ————— 0s 610us/step - accuracy: 0.9550 - loss:
0.1344
Epoch 40/100
92/92 ————— 0s 639us/step - accuracy: 0.9569 - loss:
0.1222
Epoch 41/100
92/92 ————— 0s 653us/step - accuracy: 0.9487 - loss:
0.1427
Epoch 42/100
92/92 ————— 0s 676us/step - accuracy: 0.9577 - loss:
0.1218
Epoch 43/100
92/92 ————— 0s 610us/step - accuracy: 0.9525 - loss:
0.1240
Epoch 44/100
92/92 ————— 0s 603us/step - accuracy: 0.9549 - loss:
0.1168
Epoch 45/100
92/92 ————— 0s 635us/step - accuracy: 0.9540 - loss:
0.1248
Epoch 46/100
92/92 ————— 0s 645us/step - accuracy: 0.9649 - loss:
0.1097
Epoch 47/100
92/92 ————— 0s 635us/step - accuracy: 0.9617 - loss:
0.1058
Epoch 48/100
92/92 ————— 0s 634us/step - accuracy: 0.9609 - loss:
```

```
0.1136
Epoch 49/100
92/92 _____ 0s 638us/step - accuracy: 0.9610 - loss:
0.1115
Epoch 50/100
92/92 _____ 0s 602us/step - accuracy: 0.9540 - loss:
0.1303
Epoch 51/100
92/92 _____ 0s 638us/step - accuracy: 0.9559 - loss:
0.1340
Epoch 52/100
92/92 _____ 0s 627us/step - accuracy: 0.9589 - loss:
0.1206
Epoch 53/100
92/92 _____ 0s 622us/step - accuracy: 0.9575 - loss:
0.1224
Epoch 54/100
92/92 _____ 0s 629us/step - accuracy: 0.9569 - loss:
0.1162
Epoch 55/100
92/92 _____ 0s 624us/step - accuracy: 0.9564 - loss:
0.1183
Epoch 56/100
92/92 _____ 0s 625us/step - accuracy: 0.9526 - loss:
0.1220
Epoch 57/100
92/92 _____ 0s 620us/step - accuracy: 0.9639 - loss:
0.1063
Epoch 58/100
92/92 _____ 0s 620us/step - accuracy: 0.9658 - loss:
0.0982
Epoch 59/100
92/92 _____ 0s 615us/step - accuracy: 0.9654 - loss:
0.0933
Epoch 60/100
92/92 _____ 0s 619us/step - accuracy: 0.9526 - loss:
0.1377
Epoch 61/100
92/92 _____ 0s 616us/step - accuracy: 0.9638 - loss:
0.1112
Epoch 62/100
92/92 _____ 0s 634us/step - accuracy: 0.9617 - loss:
0.1121
Epoch 63/100
92/92 _____ 0s 611us/step - accuracy: 0.9593 - loss:
0.1074
Epoch 64/100
92/92 _____ 0s 680us/step - accuracy: 0.9658 - loss:
0.1021
```

```
Epoch 65/100
92/92 _____ 0s 625us/step - accuracy: 0.9615 - loss:
0.1056
Epoch 66/100
92/92 _____ 0s 619us/step - accuracy: 0.9671 - loss:
0.0951
Epoch 67/100
92/92 _____ 0s 602us/step - accuracy: 0.9619 - loss:
0.1051
Epoch 68/100
92/92 _____ 0s 639us/step - accuracy: 0.9615 - loss:
0.1110
Epoch 69/100
92/92 _____ 0s 639us/step - accuracy: 0.9654 - loss:
0.1068
Epoch 70/100
92/92 _____ 0s 644us/step - accuracy: 0.9619 - loss:
0.1090
Epoch 71/100
92/92 _____ 0s 642us/step - accuracy: 0.9639 - loss:
0.0910
Epoch 72/100
92/92 _____ 0s 646us/step - accuracy: 0.9570 - loss:
0.1155
Epoch 73/100
92/92 _____ 0s 832us/step - accuracy: 0.9649 - loss:
0.1066
Epoch 74/100
92/92 _____ 0s 614us/step - accuracy: 0.9561 - loss:
0.1150
Epoch 75/100
92/92 _____ 0s 631us/step - accuracy: 0.9673 - loss:
0.1026
Epoch 76/100
92/92 _____ 0s 638us/step - accuracy: 0.9725 - loss:
0.0882
Epoch 77/100
92/92 _____ 0s 659us/step - accuracy: 0.9658 - loss:
0.1017
Epoch 78/100
92/92 _____ 0s 614us/step - accuracy: 0.9693 - loss:
0.0964
Epoch 79/100
92/92 _____ 0s 622us/step - accuracy: 0.9638 - loss:
0.1053
Epoch 80/100
92/92 _____ 0s 636us/step - accuracy: 0.9694 - loss:
0.0971
Epoch 81/100
```

```
92/92 _____ 0s 614us/step - accuracy: 0.9627 - loss:
0.1041
Epoch 82/100
92/92 _____ 0s 623us/step - accuracy: 0.9595 - loss:
0.1095
Epoch 83/100
92/92 _____ 0s 621us/step - accuracy: 0.9699 - loss:
0.0935
Epoch 84/100
92/92 _____ 0s 631us/step - accuracy: 0.9712 - loss:
0.0862
Epoch 85/100
92/92 _____ 0s 608us/step - accuracy: 0.9642 - loss:
0.0957
Epoch 86/100
92/92 _____ 0s 620us/step - accuracy: 0.9662 - loss:
0.0853
Epoch 87/100
92/92 _____ 0s 638us/step - accuracy: 0.9685 - loss:
0.0863
Epoch 88/100
92/92 _____ 0s 632us/step - accuracy: 0.9684 - loss:
0.0888
Epoch 89/100
92/92 _____ 0s 639us/step - accuracy: 0.9614 - loss:
0.1016
Epoch 90/100
92/92 _____ 0s 645us/step - accuracy: 0.9676 - loss:
0.0860
Epoch 91/100
92/92 _____ 0s 622us/step - accuracy: 0.9757 - loss:
0.0833
Epoch 92/100
92/92 _____ 0s 620us/step - accuracy: 0.9639 - loss:
0.1076
Epoch 93/100
92/92 _____ 0s 626us/step - accuracy: 0.9657 - loss:
0.1098
Epoch 94/100
92/92 _____ 0s 631us/step - accuracy: 0.9723 - loss:
0.0883
Epoch 95/100
92/92 _____ 0s 624us/step - accuracy: 0.9731 - loss:
0.0801
Epoch 96/100
92/92 _____ 0s 628us/step - accuracy: 0.9711 - loss:
0.0950
Epoch 97/100
92/92 _____ 0s 617us/step - accuracy: 0.9613 - loss:
```

```
0.0980
Epoch 98/100
92/92 _____ 0s 614us/step - accuracy: 0.9712 - loss:
0.0798
Epoch 99/100
92/92 _____ 0s 623us/step - accuracy: 0.9660 - loss:
0.0861
Epoch 100/100
92/92 _____ 0s 643us/step - accuracy: 0.9711 - loss:
0.0858
```

```
<keras.src.callbacks.history.History at 0x35a9fe000>
```

```
from sklearn.metrics import accuracy_score, f1_score,
balanced_accuracy_score
```

```
def compare_classification_model(model, x_train, x_val, y_train,
y_val):
```

```
    # Получаем предсказания (порог 0.5)
```

```
    y_train_pred = (model.predict(x_train) > 0.5).astype('int64')
```

```
    y_val_pred = (model.predict(x_val) > 0.5).astype('int64')
```

```
    # Вычисляем метрики для обучающей выборки
```

```
    accuracy_train = accuracy_score(y_train, y_train_pred)
```

```
    f1_train = f1_score(y_train, y_train_pred)
```

```
    # Вычисляем метрики для валидационной выборки
```

```
    accuracy_val = accuracy_score(y_val, y_val_pred)
```

```
    f1_val = f1_score(y_val, y_val_pred)
```

```
    # Выводим результаты
```

```
    print(f'Accuracy for train: {accuracy_train}')
```

```
    print(f'F1 Score for train: {f1_train}')
```

```
    print(f'Accuracy for validation: {accuracy_val}')
```

```
    print(f'F1 Score for validation: {f1_val}')
```

```
    print('-----')
```

```
# Для каждого набора данных извлекаем признаки и целевую переменную из
обучающих и валидационных выборок:
```

```
# 1. Исходный набор (d)
```

```
d_X_train = train_data.drop("Class", axis=1).values
```

```
d_Y_train = train_data["Class"].values
```

```
d_X_val = val_data.drop("Class", axis=1).values
```

```
d_Y_val = val_data["Class"].values
```

```
# 2. Стандартизированный исходный набор (sd)
```

```
sd_X_train = train_data_scaled.drop("Class", axis=1).values
```

```
sd_Y_train = train_data_scaled["Class"].values
```

```

sd_X_val    = val_data_scaled.drop("Class", axis=1).values
sd_Y_val    = val_data_scaled["Class"].values

# 3. Изменённый набор (dc)
dc_X_train = train_data_changed.drop("Class", axis=1).values
dc_Y_train = train_data_changed["Class"].values
dc_X_val    = val_data_changed.drop("Class", axis=1).values
dc_Y_val    = val_data_changed["Class"].values

# 4. Стандартизированный изменённый набор (sdc)
sdc_X_train = train_data_changed_scaled.drop("Class", axis=1).values
sdc_Y_train = train_data_changed_scaled["Class"].values
sdc_X_val    = val_data_changed_scaled.drop("Class", axis=1).values
sdc_Y_val    = val_data_changed_scaled["Class"].values

# Сравниваем результаты моделей:
print("----- FCNN на d -----")
compare_classification_model(d_fc_model, d_X_train, d_X_val,
d_Y_train, d_Y_val)

print("----- CNN на d -----")
compare_classification_model(d_cnn_model, d_X_train, d_X_val,
d_Y_train, d_Y_val)

print("----- FCNN на sd -----")
compare_classification_model(sd_fc_model, sd_X_train, sd_X_val,
sd_Y_train, sd_Y_val)

print("----- CNN на sd -----")
compare_classification_model(sd_cnn_model, sd_X_train, sd_X_val,
sd_Y_train, sd_Y_val)

print("----- FCNN на dc -----")
compare_classification_model(dc_fc_model, dc_X_train, dc_X_val,
dc_Y_train, dc_Y_val)

print("----- CNN на dc -----")
compare_classification_model(dc_cnn_model, dc_X_train, dc_X_val,
dc_Y_train, dc_Y_val)

print("----- FCNN на sdc -----")
compare_classification_model(sdc_fc_model, sdc_X_train, sdc_X_val,
sdc_Y_train, sdc_Y_val)

print("----- CNN на sdc -----")
compare_classification_model(sdc_cnn_model, sdc_X_train, sdc_X_val,
sdc_Y_train, sdc_Y_val)

----- FCNN на d -----
92/92 ————— 0s 229us/step

```


23/23 _____ 0s 234us/step
Accuracy for train: 0.953125
F1 Score for train: 0.9397379912663756
Accuracy for validation: 0.9538043478260869
F1 Score for validation: 0.9405594405594405

----- CNN на d -----

92/92 _____ 0s 333us/step
23/23 _____ 0s 374us/step
Accuracy for train: 0.9633152173913043
F1 Score for train: 0.9530434782608695
Accuracy for validation: 0.9538043478260869
F1 Score for validation: 0.9407665505226481

----- FCNN на sd -----

92/92 _____ 0s 214us/step
23/23 _____ 0s 257us/step
Accuracy for train: 0.9836956521739131
F1 Score for train: 0.9792207792207792
Accuracy for validation: 0.9538043478260869
F1 Score for validation: 0.9405594405594405

----- CNN на sd -----

92/92 _____ 0s 327us/step
23/23 _____ 0s 364us/step
Accuracy for train: 0.9884510869565217
F1 Score for train: 0.9852558542931483
Accuracy for validation: 0.9551630434782609
F1 Score for validation: 0.9430051813471503

----- FCNN на dc -----

92/92 _____ 0s 227us/step
23/23 _____ 0s 235us/step
Accuracy for train: 0.9453125
F1 Score for train: 0.9301518438177874
Accuracy for validation: 0.9429347826086957
F1 Score for validation: 0.9270833333333334

----- CNN на dc -----

92/92 _____ 0s 320us/step
23/23 _____ 0s 332us/step
Accuracy for train: 0.959578804347826
F1 Score for train: 0.9485961123110151
Accuracy for validation: 0.9483695652173914
F1 Score for validation: 0.9347079037800687

----- FCNN на sdc -----

92/92 _____ 0s 210us/step
23/23 _____ 0s 250us/step

```
Accuracy for train: 0.9755434782608695
F1 Score for train: 0.9687228496959166
Accuracy for validation: 0.9470108695652174
F1 Score for validation: 0.9319371727748691
```

```
-----
----- CNN на sdc -----
92/92 _____ 0s 330us/step
23/23 _____ 0s 460us/step
Accuracy for train: 0.9769021739130435
F1 Score for train: 0.9701230228471002
Accuracy for validation: 0.9497282608695652
F1 Score for validation: 0.9349736379613357
-----
```

```
from sklearn.metrics import accuracy_score, balanced_accuracy_score,
f1_score
```

```
# Извлекаем признаки и целевую переменную из тестового набора для
стандартизированного исходного набора (sd)
sd_X_test = test_data_scaled.drop("Class", axis=1).values
sd_Y_test = test_data_scaled["Class"].values
```

```
# Делаем предсказания с помощью лучшей модели (CNN на sd, т.е.
sd_cnn_model)
sd_Y_test_pred = (sd_cnn_model.predict(sd_X_test) >
0.5).astype('int64')
```

```
# Вычисляем метрики на тестовом наборе
accuracy_test = accuracy_score(sd_Y_test, sd_Y_test_pred)
balanced_accuracy_test = balanced_accuracy_score(sd_Y_test,
sd_Y_test_pred)
f1_test = f1_score(sd_Y_test, sd_Y_test_pred)
```

```
print("Test metrics for best model (CNN on sd):")
print(f"Accuracy: {accuracy_test}")
print(f"Balanced Accuracy: {balanced_accuracy_test}")
print(f"F1 Score: {f1_test}")
```

```
29/29 _____ 0s 1ms/step
Test metrics for best model (CNN on sd):
Accuracy: 0.9446254071661238
Balanced Accuracy: 0.9408231878906366
F1 Score: 0.9292649098474342
```

```
from scikeras.wrappers import KerasClassifier
from sklearn.model_selection import GridSearchCV
import numpy as np
```

```
# Оборачиваем функцию создания модели, используя вашу функцию
build_cnn_model.
```

```

model = KerasClassifier(
    model=build_cnn_model, # используем вашу функцию, которая
    принимает input_dim, filters, kernel_size, dense_units, dropout_rate
    verbose=0 # отключаем подробный вывод
)

# Определяем пространство гиперпараметров.
param_grid = {
    'model__filters': [32, 64],
    'model__kernel_size': [3, 5],
    'model__dense_units': [16, 32, 64],
    'model__dropout_rate': [0.3, 0.5],
    'batch_size': [16, 32],
    'epochs': [100],
    'model__input_dim': [sd_X_train.shape[1]] # Передаем размерность
    входных данных
}

# Объединяем обучающие и валидационные данные для проведения Grid
Search.
X_combined = np.concatenate((sd_X_train, sd_X_val), axis=0)
Y_combined = np.concatenate((sd_Y_train, sd_Y_val), axis=0)

# Инициализируем GridSearchCV с 3-кратной кросс-валидацией и метрикой
accuracy.
grid = GridSearchCV(estimator=model, param_grid=param_grid, cv=3,
    scoring='accuracy', verbose=1)

# Запуск подбора гиперпараметров.
grid_result = grid.fit(X_combined, Y_combined)

# Вывод лучших гиперпараметров и достигнутой точности.
print(f"Лучшие параметры: {grid_result.best_params_}")
print(f"Лучший результат (accuracy): {grid_result.best_score}")

Fitting 3 folds for each of 48 candidates, totalling 144 fits
Лучшие параметры: {'batch_size': 32, 'epochs': 100,
'model__dense_units': 64, 'model__dropout_rate': 0.3,
'model__filters': 32, 'model__input_dim': 57, 'model__kernel_size': 3}
Лучший результат (accuracy): 0.9483683019322804

from sklearn.metrics import accuracy_score, balanced_accuracy_score,
f1_score

# Используем лучшую модель из Grid Search для предсказаний на тестовом
наборе sd
sd_Y_test_pred_grid = (grid_result.best_estimator_.predict(sd_X_test)
> 0.5).astype('int64')

# Вычисляем метрики на тестовом наборе для модели из Grid Search

```

```
accuracy_grid = accuracy_score(sd_Y_test, sd_Y_test_pred_grid)
balanced_accuracy_grid = balanced_accuracy_score(sd_Y_test,
sd_Y_test_pred_grid)
f1_grid = f1_score(sd_Y_test, sd_Y_test_pred_grid)

print("Test metrics for best Grid Search model (CNN on sd):")
print(f"Accuracy: {accuracy_grid}")
print(f"Balanced Accuracy: {balanced_accuracy_grid}")
print(f"F1 Score: {f1_grid}")
```

```
Test metrics for best Grid Search model (CNN on sd):
Accuracy: 0.9457111834961998
Balanced Accuracy: 0.9417192452383067
F1 Score: 0.9305555555555556
```